

Film Cooling

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I. Introduction

The need to protect solid surfaces exposed to high-temperature environments is an old one. In general the high-temperature environment is gaseous, and it may be highly ionized as in the stream surrounding a vehicle reentering the atmosphere or in the constrictor of an electric arc or plasma jet. During the last twenty-five years, relatively sophisticated

cooling methods have been used in rockets, reentering space vehicles, high performance gas turbines, and plasma jets.

One method is to introduce a secondary fluid into the boundary layer on the surface to be protected. There are different means of introducing this secondary (or injected or coolant) fluid into the boundary layer including ablation, transpiration (or sweat or mass transfer), and film cooling. In ablation cooling, an added coating or "heat shield" decomposes, and by sublimation and other highly endothermic processes a significant quantity of gas enters the boundary layer. In transpiration cooling, the surface is usually a porous material, and the secondary fluid enters the boundary layer through this permeable surface.

Both ablation and transpiration cooling are primarily designed to protect the region where the secondary fluid enters the boundary layer. They are highly effective in this regard as a considerable portion of the heat transferred toward the wall can be taken up by the injected coolant right where the heat transfer load is highest. In addition the gas entering the boundary layer effectively thickens it, decreasing the heat transfer rate. These two methods do, however, suffer from serious disadvantages which preclude their use in many applications. The ablating material is not in general renewable and so ablation cooling has been restricted to systems with high heat fluxes of short duration, such as reentering vehicles. This restriction does not apply to transpiration cooling since a coolant can be continually introduced through the porous surface. However, porous materials to date have not had the high strength required for certain applications (e.g., turbine rotor blades) and small pore size often leads to clogging and a resulting maldistribution of coolant flow. In addition, variation in the external pressure distribution can result in a nonoptimum secondary flow distribution through the permeable surface.

Although a secondary fluid is also added to the boundary layer in film cooling there are considerable differences in operation and even in goals as compared with ablation and transpiration cooling. A key difference is that film cooling is not primarily intended as protection of the surface just at the location of coolant addition, but rather the protection of the region downstream of the injection location. Film cooling is thus the introduction of a secondary fluid (coolant or injected fluid) at one or more discrete locations along a surface exposed to a high temperature environment to protect that surface not only in the immediate region of injection but also in the downstream region.

Eckert and Livingood (*1*) examined transpiration and film cooling (as well as internal convective cooling) to see how a given amount of fluid could be used most effectively. In their comparison, however,

the maximum rather than the average temperature of a film cooled wall downstream of the injection slot was considered. As might be expected, they found transpiration cooling more efficient in use of coolant. If a real conducting wall were considered, the average film cooled wall temperature would be more appropriate, and the difference in the effectiveness of the two methods would be greatly reduced.

The geometry and flow field at the point of injection are significant variables in film cooling. In two-dimensional (including axisymmetric) film cooling not only is the external flow two-dimensional, but the secondary fluid is also introduced uniformly across the span as in Fig. 1. Secondary fluid can enter through a porous region (Fig. 1a) or through a continuous slot at some angle to the wall surface and the mainstream (Figs. 1b and 1c). The flow downstream of a transpiration

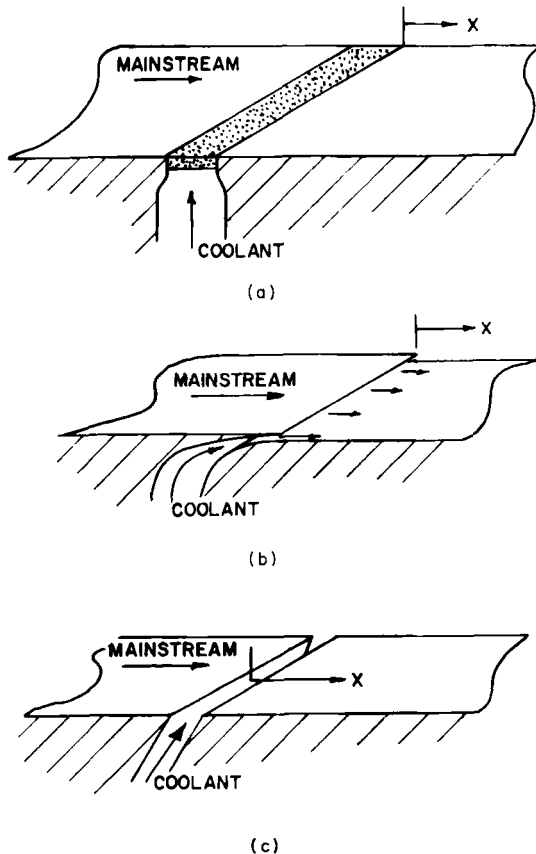


FIG. 1. Representative two-dimensional film cooling geometries: (a) porous slot, (b) tangential injection—step down slot, (c) slot angled to mainstream.

cooled or ablation cooled region is similar to film cooling, the porous wall of the transpiration cooled section or coated wall section of ablation region acting as the slot or injection region. Similarly with liquid film cooling the cooling effect in the region downstream of the point where all the liquid is evaporated can be considered similar to gas-to-gas film cooling.

Although the injection geometry can influence the film cooling performance in two-dimensional flow, the effect is usually of second order compared to geometrical effects in three-dimensional film cooling. In this latter flow (Fig. 2) the injection of secondary fluid is not uniform across the span, but rather occurs at isolated locations often through discrete holes in the surface. This can lead to the jets of secondary fluid being blown off the surface and the mainstream flow coming between and/or under the coolant jets decreasing the effectiveness of the film cooling process. Even so, for structural reasons it is usually impossible to have a truly continuous two-dimensional injection slot,

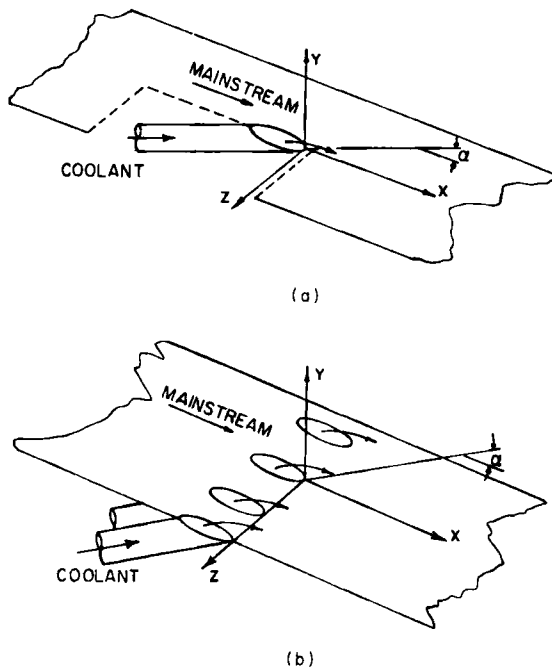


FIG. 2. Film cooling with injection through inclined tubes: (a) injection through single tube inclined at angle α to mainstream, (b) injection through single row of discrete tubes inclined at angle α to mainstream.

and so interrupted slots and even rows of multiple holes have been used.

Although film cooling has primarily been used to reduce the convective heat transfer rate from a hot gas stream to an exposed wall, it could also be used to shield a surface from thermal radiation if the radiation absorptivity of the injectant is high. This can be effectively accomplished with gas particle suspensions or a liquid coolant. In this review, however, all the fluids are considered transparent to radiation, and the convective and radiation heat transfer are then independent and can be treated separately. Only the effect of film cooling on convective heat transfer will be considered.

The introduction of secondary fluid into the boundary layer with film cooling may be considered to produce an insulating layer (film) between the wall to be protected and a gas stream flowing over it. Alternatively the injected fluid can be considered as a heat sink that effectively lowers the mean temperature in the boundary layer. As will be discussed, the secondary fluid usually serves both functions. The introduction of the secondary fluid into the boundary layer at a temperature lower than the mainstream and its resultant mixing with the fluid in the boundary layer reduces the temperature in the region downstream of injection. Note that there is usually considerable mixing of the injected fluid and the mainstream flow downstream of injection. Thus the concept of a film of secondary fluid maintaining its structure for some distance downstream and isolating the solid surface from the hot mainstream is not strictly valid, especially with a gas coolant. Although a separate discrete insulating film is not produced, injection of the secondary gas can increase the boundary layer thickness and the mass of fluid entrained into the boundary layer from the free stream. The increased boundary layer thickness tends to decrease the heat transfer to the wall. However, the increased mainstream flow entrained in the boundary layer causes increased dilution of the secondary fluid with a resulting decrease in its effectiveness as a heat sink. The significance and relative importance of these two opposing effects will be discussed subsequently.

This review is restricted to film cooling with both the mainstream fluid and the secondary fluid being gases, although not necessarily the same gas, and with a turbulent boundary layer downstream of injection. Both two-dimensional and three-dimensional secondary flow geometries are considered. In addition, film cooling in compressible flows as well as in incompressible flows is examined. Although the emphasis is on adiabatic wall temperatures and uniform mainstreams, the effects of heat transfer and variable free stream velocity are discussed.

II. Adiabatic Wall Temperature and Film Cooling Effectiveness

A. INCOMPRESSIBLE FLOW

In most film cooling applications the heat transfer from the hot gas to the surface to be protected is not zero. There is usually some type of internal cooling, or, in a transient problem, the heat capacity of the wall material itself is used to take up the heat transferred. The general problem in film cooling is to predict or measure for a given geometry, mainstream, and secondary flows the relationship between the wall temperature distribution and heat transfer. Conversely, for a given mainstream and allowable wall heat transfer the requirement may be to predict the secondary flow needed to maintain the surface temperature below some critical value.

With constant property flows the velocity distribution is independent of the temperature field and it is convenient to use the concept of a heat transfer coefficient. Thus

$$q = h \Delta T = h(T_w - T_{\text{dat}}) \quad (1)$$

where T_w is the local wall temperature. A question arises as to the datum (i.e., base or reference) temperature T_{dat} to use in Eq. (1). In the limiting case of a perfectly insulated (i.e., adiabatic) surface the heat flux would be zero and the resulting surface temperature (distribution) is called the adiabatic wall temperature T_{aw} . Thus the adiabatic wall temperature could be used as the datum temperature. The heat flux with film cooling would then be

$$q = h(T_w - T_{\text{aw}}) \quad (2)$$

Use of Eq. (2) yields a heat transfer coefficient that is independent of the temperature difference for a constant property flow. Note that in the absence of blowing, T_{aw} would be equivalent to the free stream temperature or in the case of high speed flow the recovery temperature.

Most film cooling studies have treated the determination of the heat transfer coefficient and adiabatic wall temperature distribution separately with primary emphasis on the latter. Often the heat transfer coefficient is found to be relatively close to the value without secondary flow, i.e., dependent primarily on the mainstream boundary layer flow. On the other hand, the adiabatic wall temperature distribution can vary considerably and is thus harder (and more important) to predict. In addition the adiabatic wall temperature is significant in that it is the limiting value of wall temperature that can be obtained without internal wall cooling.

Primary emphasis is given to prediction and measurement of the

adiabatic wall temperature distribution even when the assumption of a constant property flow is invalid, for example in high speed compressible flow. For this case the use of a reference temperature or reference enthalpy as with normal boundary layer flow in the absence of injection will be found useful in predicting heat transfer and will permit application of Eq. (2).

The adiabatic wall temperature is not only a function of the geometry and the primary and secondary flow fields but also the temperatures of the two gas streams. To eliminate this temperature dependence a dimensionless adiabatic wall temperature, η , called the film cooling effectiveness is used. For low speed, constant property flow the film cooling effectiveness is given by

$$\eta = \frac{T_{aw} - T_{\infty}}{T_2 - T_{\infty}} \quad (3)$$

where the temperatures of the secondary fluid T_2 and the mainstream fluid T_{∞} are assumed constant. Note that in general $T_{aw} < T_{\infty}$ and $T_2 < T_{\infty}$ in a film cooling application. Since the constant property energy equation is linear in temperature the film cooling effectiveness is dependent only on the primary and secondary flows and the position on the surface. Note that the film cooling effectiveness usually varies from unity at the point of injection (where $T_{aw} = T_2$) to zero far downstream where, because of dilution of the secondary flow, the adiabatic wall temperature approaches the free stream temperature.

B. HIGH-SPEED FLOW

For high-speed flow the film cooling effectiveness must be defined somewhat differently. At the point of injection the wall temperature T_{w2} would be expected to be the recovery temperature of the secondary flow or possibly the total temperature of the secondary flow. Far downstream the wall temperature might be expected to approach the mainstream recovery temperature T_r (evaluated in the absence of secondary flow). An expression often used for compressible flow film cooling is

$$\eta_r = \frac{T_{aw} - T_r}{T_{w2} - T_r} \quad (4)$$

Note that the film cooling effectiveness η_r reduces to Eq. (3) when compressibility effects can be neglected.

An alternate convenient definition of effectiveness for high speed flows employs the isoenergetic-injection wall temperature distribution,

$$\eta_{is} = \frac{T_{aw} - T_{awi}}{T_{w2} - T_{w2i}} \quad (5)$$

The isoenergetic temperatures (T_{aw_i} and T_{w2i}) are the temperatures on the adiabatic wall for the same mainstream conditions, and the same secondary flow rate, but with secondary flow stagnation temperature equal to the stagnation temperature of the mainstream (see Fig. 3). Note that T_{aw_i} is a function of position while the recovery temperature used in Eq. (4) can often be considered constant. The definition of effectiveness (Eq. 5) as in Eqs. (3) and (4) gives an effectiveness of unity at the point of injection and a zero value of effectiveness far downstream.

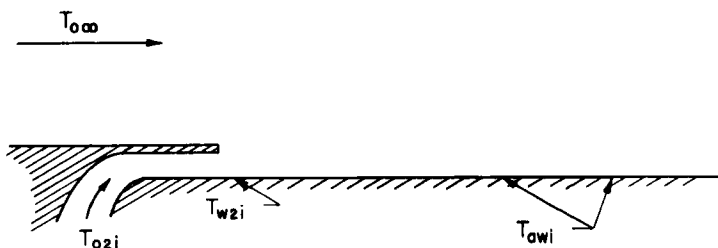


FIG. 3. Temperatures with isoenergetic injection in high speed flow, injection stagnation temperature T_{02i} equal to the mainstream stagnation temperature $T_{0\infty}$.

The isoenergetic film cooling effectiveness is found to correlate experimental results better than an effectiveness based on the recovery temperature without secondary flow. The isoenergetic injection temperature distribution T_{aw_i} is obtained at the same blowing rate and free stream conditions as with film cooling. Assuming a constant property fluid, the flow fields are the same in the two cases and independent of the temperature of the secondary fluid. The viscous dissipation terms are also the same in the isoenergetic and normal film cooling runs and subtraction of the temperature distributions from the two runs eliminates the viscous dissipation effect. Thus the difference solution η_{is} for the high-speed flow should be the same as for low-speed flows if the pertinent dimensionless variables describing the flow remain the same. For low-speed flow η_{is} reduces to the usual incompressible flow effectiveness (Eq. 3). For these reasons it may be possible to predict the isoenergetic film cooling effectiveness using the results (theoretical or experimental) from low-speed flows.

When using compressible-flow film cooling, results in a design problem not only η_{is} but also the adiabatic wall temperature distribution for isoenergetic injection T_{aw_i} must be known. However, for many applications the difference between the recovery temperature without

secondary flow, T_r , and T_{awi} is much smaller than the difference between either of these and the expected temperature of the secondary fluid. Then the recovery temperature T_r may be used as a reasonable approximation for T_{awi} .

The enthalpy can also be used in defining the film cooling effectiveness in high-speed flows. This would be useful at large temperature differences. The proper enthalpies could then be substituted (for the temperatures) directly into the above expressions for effectiveness.

C. IMPERMEABLE WALL CONCENTRATION

It is often difficult to design test systems with walls that sufficiently approximate adiabatic surfaces. This is particularly apparent when the adiabatic wall temperature distribution has large gradients and with a very high temperature mainstream. In such cases a mass transfer process can be used as an analogue to film cooling. Thus instead of injecting a gas at a different temperature from the mainstream, a gas of different composition would be injected isothermally. This might apply particularly in a study of the effects of three-dimensional film cooling with large density differences (between the primary and secondary flows). The mass transfer analogy is also useful for two-dimensional film cooling with large temperature (density) differences. The injected gas can be completely different in composition from the free stream, or only a tracer gas might be used in the secondary flow. If the secondary fluid is otherwise the same as the mainstream, the use of a tracer gives results comparable to low density (temperature) differences.

The mass transfer process is analogous to the heat transfer process (neglecting thermal diffusion phenomena) if the equivalent dimensionless parameters of the flow are the same in the two cases and if the Lewis number is unity. (The Lewis number is the ratio of the Schmidt number for the mass transfer process to the Prandtl number for the corresponding heat transfer process.) The turbulent Lewis number as well as the molecular Lewis number should be unity for the analogy to hold. If the flow is sufficiently turbulent, variations in the molecular Lewis number from unity may not play an important role, but in all cases studied to date the value of the turbulent Lewis number should be considered.

When using the mass transfer analogy with foreign gas injection, the quantity analogous to the adiabatic wall temperature is the concentration of the injected gas at an impermeable wall. Although there is some question as to the proper concentration to use, the mass fraction C is the most widely used. Equivalent to the film cooling effectiveness

based on adiabatic wall temperature is an effectiveness based on the impermeable wall concentration

$$\eta_c = \frac{C_{iw} - C_\infty}{C_2 - C_\infty} \quad (6)$$

If the secondary fluid contains a single constituent not contained in the mainstream, then $C_\infty = 0$, $C_2 = 1$ and Eq. (6) becomes

$$\eta_c = C_{iw} \quad (7)$$

III. Analysis

A. GENERAL REMARKS

A number of theoretical correlations and predictions have been developed for the film cooling effectiveness. Since interest is chiefly in turbulent film cooling, the analyses are at least partly empirical yet often suggestive of the significant features of the flow. Much of the interest has centered on relatively simple heat sink models in which the added secondary flow is considered as a sink of heat at the point of injection reducing the temperature in the downstream boundary layer and thus the temperature of the wall. The models have been applied to two-dimensional incompressible and compressible flows and lately to three-dimensional film cooling. Other analyses use some of the recent numerical techniques for predicting two-dimensional turbulent boundary layers and separated flows to obtain predictions of film cooling effectiveness.

In this section some of the theoretical analyses will be developed and differences between them discussed. Comparison with experimental results will be deferred till the next section in which the various experimental studies are described.

B. TWO-DIMENSIONAL INCOMPRESSIBLE FLOW FILM COOLING—HEAT SINK MODEL

The first heat sink model for film cooling was given by Tribus and Klein (2) in a work primarily concerned with developing kernels to predict the heat transfer and temperature distribution along non-isothermal surfaces. At the suggestion of Eckert, they used Rubesin's kernel for the wall temperature distribution with a turbulent boundary layer to predict the temperature on an adiabatic surface downstream from a line source of heat. The strength of the line source is determined

by the net enthalpy flow of the secondary fluid $|\rho_2 U_2 C_{p2}(T_2 - T_\infty)|$ and the calculation ignores any effect of the injection on the mainstream flow. Integration of the kernel yields

$$\begin{aligned}\eta &= 5.76 \text{Pr}^{2/3} \text{Re}_2^{0.2} (\mu_2/\mu_\infty)^{0.2} (C_{p2}/C_{p\infty})(x/Ms)^{-0.8} \\ &= 5.76 \text{Pr}^{2/3} (C_{p2}/C_{p\infty}) \xi^{-0.8}\end{aligned}\quad (8)$$

where

$$\xi = (x/Ms)[(\mu_2/\mu_\infty) \text{Re}_2]^{-0.25} \quad (9)$$

The dimensionless blowing rate parameter M is the ratio of the mass velocity of the injected fluid to the mainstream mass velocity. The distance x is measured downstream from the point of injection. Tribus and Klein compare their analysis to results for air injected into an air mainstream. For $\mu_2 = \mu_\infty$, $C_{p2} = C_{p\infty}$, and $\text{Pr} \approx 0.72$,

$$\eta = 4.62 \text{Re}_2^{0.2} (x/Ms)^{-0.8} \quad (10)$$

The parameters in Eq. (8) appear in essentially all of the heat sink models and they are very useful for correlating data, particularly at low blowing rates.

Equations (8) and (10), however, predict higher values of effectiveness than have usually been found experimentally. This is apparently due to the assumption that the injected gas does not affect the velocity boundary layer. Figure 4, however, indicates that the boundary layer is considerably thickened by injection. This particular figure was obtained for secondary flow through a porous section (3).

Later heat sink analyses were made by Librizzi and Cresci (4), Kutateladze and Leont'ev (5), Stollery and El-Ehwany (6, 7), and Goldstein and Haji-Sheikh (8). These analyses have a great many similarities, and all proceed from an initial energy balance on the boundary layer.

C. ENERGY BALANCE IN THE BOUNDARY LAYER

The mass flowing within the boundary layer is considered to be composed of two different fluids from two different streams—the injected gas ($\dot{m}_2$) and the mass which enters (is entrained into) the boundary layer from the mainstream ($\dot{m}_\infty$). These gases are assumed to be well mixed in the boundary layer. At any position downstream of injection the mass flowing per unit time ($\dot{m}$) in the boundary layer is given by (see Fig. 5a)

$$\dot{m} = \dot{m}_2 + \dot{m}_\infty = \int_0^\delta \rho U dy \quad (11)$$

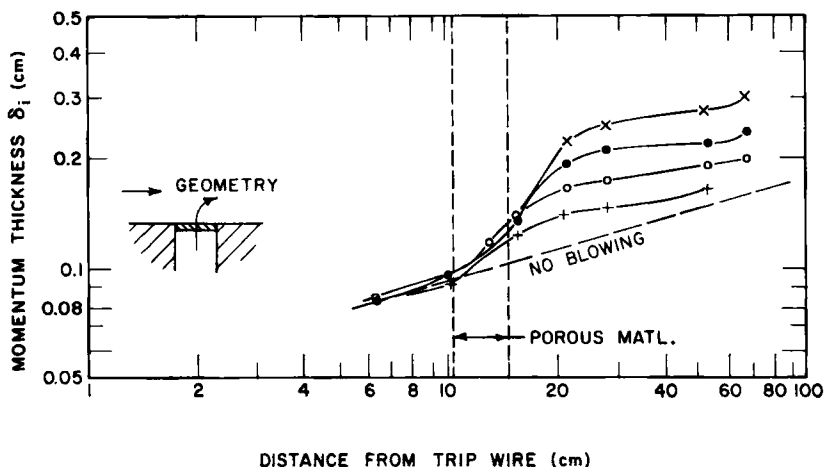


FIG. 4. Effect of injection on boundary-layer momentum thickness for injection through a porous slot.

Symbol	M	$\dot{m}_2(\text{gm/sec})$	$U_\infty(\text{m/sec})$
×	0.0389	22.55	56.1
○	0.0263	15.41	55.2
●	0.0162	10.08	54.6
+	0.00776	4.90	56.5

[R. J. Goldstein, G. Shavit, and T. S. Chen, *J. Heat Transfer* 87, 353 (1965).]

The mean temperature $\bar{T}$ in the boundary layer is given by

$$\bar{T} - T_\infty = \frac{\int_0^{\delta} \rho U C_p (T - T_\infty) dy}{\int_0^{\delta} \rho U C_p dy} \quad (12)$$

Assume constant property ideal gases. The average specific heat $\bar{C}_p$ for the boundary layer is given by,

$$\bar{C}_p = (\dot{m}_2 C_{p2} + \dot{m}_\infty C_{p\infty}) / (\dot{m}_2 + \dot{m}_\infty) \quad (13)$$

If the wall over which the fluid flows is adiabatic, application of the steady flow energy equation (see Fig. 5b) at any downstream position yields

$$(\dot{m}_2 + \dot{m}_\infty) \bar{C}_p \bar{T} = \dot{m}_2 C_{p2} T_2 + \dot{m}_\infty C_{p\infty} T_\infty \quad (14)$$

Rearranging terms in Eq. (14) and using Eq. (13) yields

$$\frac{\bar{T} - T_\infty}{T_2 - T_\infty} = \frac{1}{\left(1 + \frac{\dot{m}_\infty C_{p\infty}}{\dot{m}_2 C_{p2}}\right)} \quad (15)$$

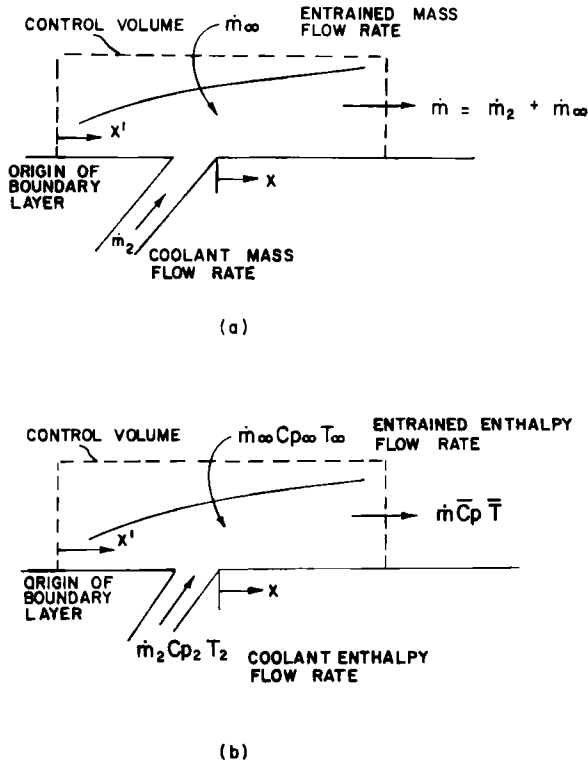


FIG. 5. Control volume when performing (a) mass and (b) energy balances.

In References (4-6) the mean temperature $\bar{T}$ in the boundary layer is assumed equal to the adiabatic wall temperature. (Librizzi and Cresci (4) did consider the temperature variation across the boundary layer in a compressible flow model.) Equation (15) reduces to Eq. (3), and the film cooling effectiveness becomes

$$\eta = \frac{1}{\left(1 + \frac{\dot{m}_\infty C_{p\infty}}{\dot{m}_2 C_{p2}}\right)} \quad (16)$$

These analyses all use essentially the same method to predict $\dot{m}_\infty$. They assume a $\frac{1}{7}$ th power turbulent velocity profile and a boundary layer thickness given by

$$\delta/x' \approx 0.376 \text{Re}_x^{-1/5} \quad (17)$$

to predict the entrained flow rate $\dot{m}_\infty$. The parameter x' is the distance from the point at which the boundary layer starts. Some of the original derivations use a slightly different value of the constant (0.376), but for

comparison purposes they have all been recalculated using Eq. (17). Equation (17) is valid in the absence of injection and the analyses assume that it is still valid with injection. The primary difference between the three analyses is the assumed location where the total mass flow in the boundary layer starts (i.e., where $x' = 0$). For a $\frac{1}{2}$ th power velocity profile starting at $x' = 0$, the mass in the boundary layer from the mainstream at some distance downstream is

$$\dot{m}_\infty = \frac{7}{8}\rho_\infty U_\infty \delta = 0.329\rho_\infty U_\infty x' \text{Re}_{x'}^{-1/5} \quad (18)$$

Librizzi and Cresci (4) assume that the boundary layer starts at the point of injection ($x' = x$) and at injection ($x = 0$)

$$\dot{m} = \dot{m}_2 + \dot{m}_\infty = \dot{m}_2 \quad (19)$$

Using $\dot{m}_2 = \rho_2 U_2 s$

$$\dot{m}_\infty/\dot{m}_2 = 0.329(x/Ms)^{0.8} [\text{Re}_2(\mu_2/\mu_\infty)]^{-0.2} \quad (20)$$

Putting this into Eq. (16) and using Eq. (9),

$$\eta = \frac{1}{1 + 0.329(C_{p\infty}/C_{p2}) \xi^{0.8}} \quad (21)$$

For $C_{p\infty} = C_{p2}$

$$\eta = \frac{1}{1 + 0.329\xi^{0.8}} \quad (22)$$

Kutateladze and Leont'ev (5) assume the boundary layer downstream of injection grows as if it had started upstream of injection at some distance x'' . In order to calculate the distance, x'' , they assume that the upstream (fictitious) boundary layer grows as a turbulent boundary layer which started sufficiently far upstream to have a net mass flow in it at the point of injection equal to the secondary mass flow rate. Calculation of the mass flow rate from Eqs. (11) and (18) using the above assumptions gives

$$\dot{m}_\infty/\dot{m}_2 = 0.329(4.01 + \xi)^{0.8} - 1 \quad (23)$$

Inserting this into Eq. (16) gives

$$\eta = \frac{1}{1 + (C_{p\infty}/C_{p2})[0.329(4.01 + \xi)^{0.8} - 1]} \quad (24)$$

and for $C_{p2} = C_{p\infty}$

$$\eta = \frac{1}{1 + 0.249\xi^{0.8}} \quad (25)$$

Leont'ev (9) extended the model to include the effect of film cooling

on heat transfer and also the effect of rough surfaces on film cooling performance.

Stollery and El-Ehwany (6, 7) assume the boundary layer starts at injection ($x = x'$) and also that the total mass flow in the boundary layer is zero at the point of injection. Thus at $x = 0$,

$$\dot{m} = 0 \quad (26)$$

and for $x > 0$

$$\dot{m} = \frac{7}{8} \rho_{\infty} U_{\infty} \delta \quad (27)$$

Thus using Eq. (11)

$$\dot{m}_{\infty} = \frac{7}{8} \rho_{\infty} U_{\infty} \delta - \dot{m}_2 \quad (28)$$

Inserting this into Eq. (16) (assuming $C_{p2} = C_{p\infty}$) and noting $\dot{m}_2 = \rho_2 U_2 s$, yields

$$\eta = \frac{1}{\frac{7}{8} \frac{\rho_{\infty} U_{\infty} \delta}{\dot{m}_2}} = \frac{8}{7} \frac{\rho_2 U_2 s}{\rho_{\infty} U_{\infty} \delta} \quad (29)$$

Using Eq. (17)

$$\eta = 3.03(x/Ms)^{-0.8} \text{Re}_2(\mu_2/\mu_{\infty})^{0.2} \quad (30)$$

or introducing Eq. (9)

$$\eta = 3.03\xi^{-0.8} \quad (31)$$

They also indicate in their analysis the effect of foreign gas injection (when $C_{p2} \neq C_{p\infty}$) and suggest an approach for determining the film cooling effectiveness with a variable free stream velocity.

Since the above heat sink models assume complete mixing of the secondary fluid in the mainstream boundary layer, their validity would be expected to suffer when applied near the point of injection. Both the Tribus and Klein prediction (Eq. 10) and the Stollery and El-Ehwany prediction (Eq. 31) essentially assume no mass flow in the boundary layer at injection yet a finite heat source. Consequently they predict a value of infinity for the effectiveness at $x = 0$. The Librizzi and Cresci and the Kutateladze and Leont'ev correlations, by their assumption of $\dot{m} = \dot{m}_2$ at the point of injection yield an effectiveness of unity at $x = 0$. This is a convenience which should not be overlooked. Note that three of the correlations (Eqs. (22), (25), and (31)) approach the same prediction far downstream where ξ is large.

It is of interest that these last three correlations (Eq. (22), (25), and (31)) are in better agreement with experimental data, as will be shown below, than the Tribus and Klein correlation (Eq. (8) or (10)), even though they use the additional major assumption that the temperature in the boundary layer is constant. That this latter assumption is

not true was known from the earliest test results of Wieghardt (10) as is shown in Fig. 6. The temperature profile is observed to be similar for different positions downstream of injection.

The reason the correlations (Eq. (22), (25), and (31)) work so well follows from the unwritten law that sometimes two invalid assumptions are better than one. Thus the assumption that the boundary layer is unaffected by the secondary flow would indicate less flow into the boundary layer from the mainstream (i.e., less dilution) than actually occurs, reducing $\dot{m}_\infty$ in Eqs. (11) and (16), and thus predict a larger effectiveness than occurs. However, the mean temperature in reality is significantly different from the adiabatic wall temperature; $\bar{T}$ being between T_{aw} and T_∞ . Thus the assumption used to get Eq. (16) that

$$\eta = \frac{\bar{T} - T_\infty}{T_2 - T_\infty} \quad (32)$$

gives a lower effectiveness than the true value

$$\eta = \frac{T_{aw} - T_\infty}{T_2 - T_\infty} \quad (33)$$

since $|T_{aw} - T_\infty| > |\bar{T} - T_\infty|$. The success of the correlations is apparently due to these two effects counterbalancing each other.

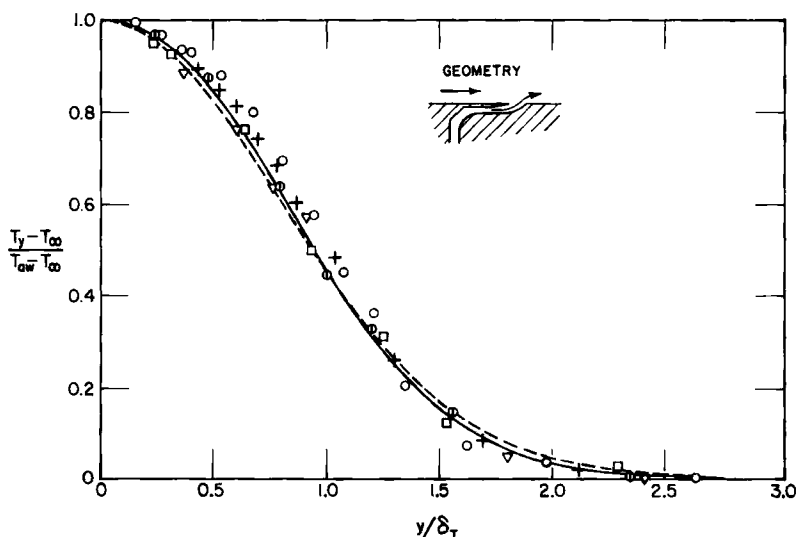


FIG. 6. Dimensionless boundary layer temperature profiles at various positions downstream of injection: $M = 0.74$, $s = 10$ mm, $T_2 - T_\infty = 28^\circ\text{C}$, (—) $\exp[-0.768(y/\delta_\tau)^{13/6}]$, (---) $\exp[-0.785(y/\delta_\tau)^2]$. Distance from injection in meters: ($\circ$) 0.05, (+) 0.2, ($\diamond$) 1.0, (∇) 2.0, ($\square$) 4.0. [K. Wieghardt, AAF Translation No. F-TS-919-RE (1946).]

Goldstein and Haji-Sheikh (8) use a modified heat sink analysis as an attempt to correct the two assumptions mentioned above. An overall energy balance is performed yielding Eq. (15). The temperature variation through the boundary layer is considered, as is the effect of the injection on increasing the size of the boundary layer, and thus the mass flow entering the boundary layer from the mainstream.

Assuming a power law velocity profile and a similar temperature profile (cf. Fig. 6) the mean temperature in the boundary layer is calculated from

$$\bar{T} - T_{\infty} = \lambda(T_{aw} - T_{\infty}) \quad (34)$$

Combining Eqs. (34) and (15),

$$\eta = \frac{1/\lambda}{1 + \frac{\dot{m}_{\infty} C_{p\infty}}{\dot{m}_2 C_{p2}}} \quad (35)$$

where λ depends on the temperature and velocity profiles. Using the profiles of Wieghardt (10) and extrapolating the results to zero blowing rate so they can be compared to Tribus and Klein results gives

$$1/\lambda = 1.9 \text{Pr}^{2/3} \quad (36)$$

though the variation with Prandtl number would not be expected to hold over a large range.

The ratio of the mass flow in the boundary layer with blowing $\dot{m}_{\infty}$, to the mass flow without blowing, $\dot{m}_{\infty 0}$, is determined from experimental results of previous investigations. Figure 7 shows that the mass added to the boundary layer from the free stream increases with secondary flow rate and angle of injection (from the mainstream direction). This figure specifically refers to the flow when the secondary and primary gases are the same. For different gases

$$\dot{m}_{\infty}/\dot{m}_{\infty 0} = 1 + 1.5 \times 10^{-4} \text{Re}_2(\mu_2 W_{\infty}/\mu_{\infty} W_2) \sin \alpha \quad (37)$$

is obtained where α is the angle of injection (measured relative to the wall). Combining Eq. (37) with Eqs. (35) and (36) and using Eq. (18) to predict $\dot{m}_{\infty 0}$ (not $\dot{m}_{\infty}$ directly),

$$\eta = \frac{1.9 \text{Pr}^{2/3}}{1 + 0.329(C_{p\infty}/C_{p2}) \xi^{0.8} \beta} \quad (38)$$

where

$$\beta = 1 + 1.5 \times 10^{-4} \text{Re}_2(\mu_2 W_{\infty}/\mu_{\infty} W_2) \sin \alpha \quad (39)$$

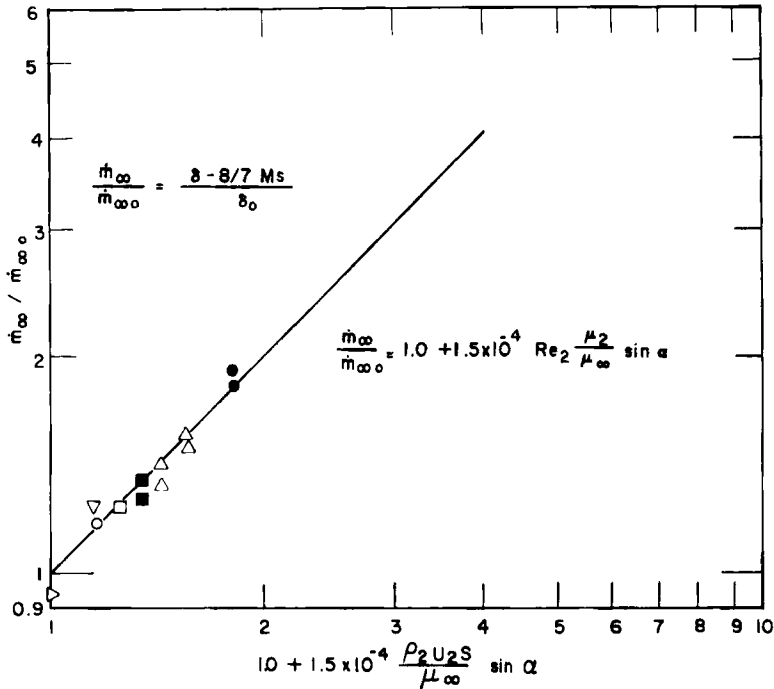


FIG. 7. Ratio of boundary layer entrained mass flow rate with secondary injection $\dot{m}_{\infty}$ to entrained mass flow rate for zero injection $\dot{m}_{\infty 0}$ as a function of secondary flow injection angle and flow rate.

Data: (○, ●, ▲, ■), R. J. Goldstein, G. Shavit, and T. S. Chen, *J. Heat Transfer* 87, 353 (1965); (□, △), K. Wieghardt, AAF Translation No. F-TS-919-RE (1946); (▽), J. P. Harnett, R. C. Birkebak, and E. R. G. Eckert, *J. Heat Transfer* 83, 293 (1961); (▷), R. A. Seban and L. H. Back, *J. Heat Transfer* 84, 45 (1962).

[R. J. Goldstein and A. Haji-Sheikh, in Japan Soc. Mech. Engr. 1967 Semi-Intern. Symp., 213-218, Tokyo (1967).]

D. TWO-DIMENSIONAL INCOMPRESSIBLE FLOW FILM COOLING—OTHER MODELS

Hatch and Papell (11) use a theoretical model for tangential injection, in which they envision that the injected gas remains in a separate film apart from the free stream and try to calculate the heat exchange between this film and a turbulent boundary layer atop it. It would be expected that such a correlation would fit best very close to the region of injection.

Saarlal (12) uses a boundary layer model to predict film cooling effectiveness. The analysis also permits an approximate calculation of the heat transfer with film cooling and the effect of a variable mainstream velocity.

Seban and Back (13, 14, 15) use the similarity of the temperature profiles to predict film cooling effectiveness based on a uniform eddy diffusivity across the boundary layer. They divide the flow (tangential injection) into three regions: a wall-jet-like flow near the slot where the secondary flow is preserved, a mixing region, and finally a normal turbulent boundary layer region. They observe (for $M < 0.8$) that the initial region of the flow is defined by $x/s < 56M^{1.5}$ (14). The model uses a linearized form of the energy equation and an upstream effective starting point of the boundary layer and predicts reasonable values of effectiveness for tangential secondary flow injection.

The wall jet region is particularly significant at a velocity ratio, U_2/U_∞ , greater than unity. Note that it is apparently the velocity ratio that plays the key role in determining the approach to wall jet behavior rather than the mass velocity ratio or blowing rate M . The significance of the velocity ratio might be expected since it indicates (for tangential injection) whether the secondary fluid will tend to accelerate the mainstream ($U_2 > U_1$) or be accelerated by the mainstream ($U_2 < U_1$).

Spalding (16) proposes relations for film cooling with a tangential slot through which a fluid similar to the mainstream fluid is injected. Although basically empirical the model reduces to a relation similar to the heat sink models at low blowing rates and behaves similar to what would be expected for a wall jet at large blowing rates. He predicts, for $\xi' < 7$,

$$\eta = 1 \quad (40)$$

and for $\xi' \geq 7$,

$$\eta = 7/\xi' \quad (40a)$$

where

$$\xi' = 0.91(x/Ms)^{0.8} \text{Re}_2^{-0.2} + 1.41\{[1 - (U_\infty/U_2)] x/s\}^{0.5} \quad (40b)$$

All of the analyses described so far use models that employ considerable empirical input. In an attempt to use a more analytical approach Whitelaw and co-workers (17, 18, 19) have tried to solve the turbulent flow boundary layer equations for film cooling with tangential injection. They use the Patankar and Spalding (20) approach in which a mixing length and effective turbulent Prandtl number distribution are assumed. Although this method has many difficulties and there are questions about its validity and accuracy, it does offer the hope of future solutions valid for the region close to injection point as well as predicting the heat transfer coefficient. Other references using this approach include (21)–(23).

E. TWO-DIMENSIONAL FILM COOLING IN A HIGH-SPEED FLOW

Film cooling in a two-dimensional high-speed flow has been analyzed (24), where the reference temperature (enthalpy) method was combined with some of the incompressible flow analyses to obtain the film cooling effectiveness. As mentioned above, an effectiveness using the isoenergetic wall temperature distribution as a reference appears to work best for compressible flows. The reference temperature used is

$$T_* = T_\infty + 0.72(T_r - T_\infty) \quad (41)$$

and all properties in the boundary layer are evaluated at this temperature. Thus

$$\xi_* = (x/Ms)(\text{Re}_2 \mu_2/\mu_*)^{-0.25} (\rho_*/\rho_\infty) \quad (42)$$

The local wall temperature T_w or T_{aw} would be used in place of T_r in Eq. (41) if very large temperature differences are encountered. Corresponding to the Kutateladze and Leont'ev model (Eq. 24)

$$\eta_{is} = \{1 + (C_{p\infty}/C_{p2})[0.33(4.00 + \xi_*)^{0.8} - 1]\}^{-1.0} \quad (43)$$

When the injected fluid is the same as the mainstream fluid the relations derived for high-speed flow are:

Kutateladze and Leont'ev model;

$$\eta_{is} = (1 + 0.25\xi_*)^{-0.8} \quad (44)$$

Librizzi and Cresci model;

$$\eta_{is} = (1 + 0.33\xi_*^{0.8})^{-1} \quad (45)$$

Stollery and El-Ehwany model;

$$\eta_{is} = 3.03\xi_*^{-0.8} \quad (46)$$

and Goldstein and Haji-Sheikh model;

$$\eta_{is} = 1.9 \text{Pr}^{2/3}(1 + 0.33\xi_*^{0.8}\beta)^{-1} \quad (47)$$

where

$$\beta = 1 + 1.5 \times 10^{-4} \text{Re}_2(\mu_2/\mu_*) \sin \alpha \quad (48)$$

In deriving these expressions for high-speed film cooling the constant used in the boundary layer growth equation comes from a best fit to experimental skin friction data proposed by Schlichting (25).

Laganelli (26) uses a similar analysis based on reference properties to predict film cooling performances in supersonic flow. His results are similar to those given above. He also extends his results to an

axisymmetric coordinate system. Librizzi and Cresci (4) have also considered film cooling in an axisymmetric supersonic flow.

F. INJECTION THROUGH DISCRETE HOLES—THREE-DIMENSIONAL FILM COOLING

The heat sink concept has been applied to film cooling following injection through discrete holes (27). With such a geometry there is little hope of getting a relatively exact analytic description of the velocity and temperature distributions.

At relatively low mass injection rates the mass addition through a single hole can be considered to act as a localized heat sink on the film-cooled surface. The transfer process in the boundary layer is approximated by the conduction equation, the problem being equivalent to determining the temperature distribution in a semi-infinite solid medium along whose surface a point heat source is moving in a straight line with constant velocity. The medium is the mainstream gas, the strength of the source is determined from the net enthalpy flow added through the hole and the velocity of the source in that of the free stream, though in the reverse direction. A major difficulty (and approximation) is to evaluate an effective thermal conductivity or thermal diffusivity of the mainstream.

The resulting temperature distribution in the mainstream is,

$$\frac{T_{aw}(X, Y, Z) - T_{\infty}}{T(0, 0, 0) - T_{\infty}} = \frac{MU_{\infty}D}{8\epsilon(X/D + 0.5)} \exp \left[-0.693 \left[\left(\frac{Y}{Y_{1/2}} \right)^2 + \left(\frac{Z}{Z_{1/2}} \right)^2 \right] \right] \quad (49)$$

Along the adiabatic surface $Y = 0$,

$$\eta(X, Z) = \frac{MU_{\infty}D}{8\epsilon(X/D + 0.5)} \exp \left[-0.693 \left(\frac{Z}{Z_{1/2}} \right)^2 \right] \quad (50)$$

The reference coordinates used in Eq. (49) are shown in Fig. 2; Y is the vertical distance from the surface, Z is the lateral distance from the center plane of the injection hole, and X is the distance downstream from injection. The distances $Y_{1/2}$ and $Z_{1/2}$ are distances at which the temperature difference drops to half its value along the centerline on the tunnel surface. Note that the effectiveness is now a function of lateral position from the hole centerline as well as of downstream location. The form of this relation is found to approximate the experimental data at low blowing rates and thus it has proven useful. Extension to higher injection rates faces formidable obstacles, particularly as the jet appears to leave the surface at large blowing rates.

Either Eq. (50) or the direct experimental results for single hole injection can be used to predict the film cooling performance of a row of holes using the principle of superposition. As long as the flows from the individual jets do not interact greatly, superposition appears to work reasonably well (28). At large blowing rates and far downstream the jets come together and superposition of single hole results to predict film cooling from a number of holes cannot be used.

IV. Experimental Studies

A. GENERAL REMARKS

A summary of some of the experimental studies of film cooling is presented in Table I. A brief description of the geometry of secondary injection as well as the range of pertinent experimental variables is included. Discussion of the results of the individual investigations is given below.

In many of the tests, film "heating" rather than film cooling is employed for reasons of convenience. With small temperature differences the flow can be considered constant property. Then, if there is no radiant energy input, the dimensionless temperature distribution in the boundary layer (and thus the film cooling effectiveness) will be independent of whether the secondary gas is hotter or colder than the mainstream. This has been discussed previously in the section on film cooling effectiveness and is also applicable to the determination of the heat transfer coefficient when a heated or cooled wall is used.

In some applications, however, there can be a considerable temperature difference between the mainstream and the coolant. Then a key parameter may be the ratio of the densities of the two fluids. This can, of course, be studied using real temperature differences, but attainment of adiabatic wall conditions is very difficult with large temperature differences, so several investigations have utilized the mass transfer analogy. The equivalent of the film cooling effectiveness is then the impermeable wall effectiveness as discussed above. This method is convenient even at small temperature (density) differences as there are always errors introduced due to thermal conduction in the wall when finite temperature differences are employed. This is particularly true in three-dimensional film cooling though the analogy has not been applied there as yet. One difficulty in using the mass transfer analogy is the simulation of a (heat) transfer process at the wall-gas interface. So far only impermeable surfaces have been used. There is also the lingering question of the direct equivalence of the analogy; Burns and Stollery (29), in particular, question whether the turbulent Lewis number is unity.

B. TWO-DIMENSIONAL FILM COOLING—INCOMPRESSIBLE FLOW

1. *Injection of Air into Air—Constant Mainstream Velocity*

The first well-known study, Wieghardt (10), on film cooling not only used a heated secondary gas, but was specifically applied to a film-heating problem, namely, the de-icing of airplane wings. Air was ejected through a slot (see Table I) inclined at an angle of about 30° to the wall surface. Note that as in most studies in which injection occurs through a slot the secondary flow was not fully developed nor was the temperature profile completely uniform at the slot exit. As previously noted, Wieghardt found that the temperature profiles were approximately similar and could be expressed (Fig. 6) in the form

$$\frac{T_y - T_\infty}{T_{aw} - T_\infty} \sim \exp[-0.768(y/\delta_T)^{13/6}] \quad (51)$$

Some distance downstream of injection, temperature distributions similar to this have been found in other film cooling studies indicating the relative insensitivity of the temperature profile to the secondary injection geometry. At high blowing rates and near the injection location, however, similarity of the temperature profile may not be assumed.

Wieghardt found a maximum in film cooling effectiveness at a blowing rate of about unity (Fig. 8). He was able to correlate his adiabatic wall temperature distribution at low blowing rates ($M < 1$) and some distance downstream ($x/s > 100$) with a simple relation

$$\eta = 21.8[x/Ms]^{-0.8} \quad (52)$$

which, since his range of slot Reynolds number Re_s was small, is not too different from some of the predicted results of the heat sink model. This simple equation, even today, is widely used to obtain initial estimates of film cooling performance.

Other workers (30), using the same geometry as Wieghardt, found a relation similar to Eq. (52) although the numerical value of the constant was lower (16.9 instead of 21.8). Eckert and Birkebak (31) using the same geometry were able to correlate their results with Eq. (52).

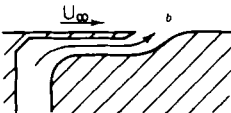
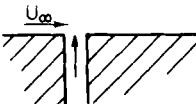
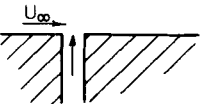
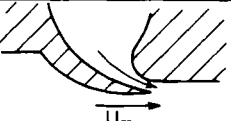
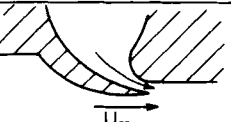
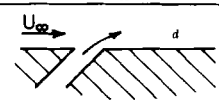
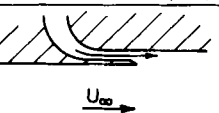
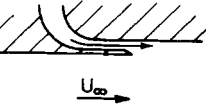
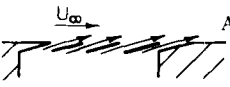
Others (32) studied film cooling with injection through both a normal and a tangential slot. For normal injection they found

$$\eta = 2.2(x/Ms)^{-0.5} \quad (53)$$

Seban (33) studying the film cooling effectiveness downstream of a stepdown slot (Fig. 1b) correlated his data at low blowing rate ($M < 1$) with the relation

$$\eta = 25M^{0.4}(x/Ms)^{-0.8} \quad (54)$$

TABLE 1

Ref.	Geometry	Injection gas ^a	Density ratio, ρ_2/ρ_∞	Temp. ratio, T_2/T_∞	Blowing rate, $M = \rho_2 U_2 / \rho_\infty U_\infty$	Free stream Mach No.
Wieghardt (10)		Air	0.78 to 0.91	1.1 to 1.2	0.22 to 1.9 + wall jet	0.046 to 0.092
Scesa (60)		Air	0.81 to 0.91	1.09 to 1.23	0.2 to 1.14	0.025 to 0.065
Seban, Chan Scesa (32)		Air	0.81 to 0.92	1.09 to 1.23	0.08 to 0.916	0.037 to 0.108
		Air	0.81 to 0.92	1.09 to 1.23	0.19 to 1.14	0.025 to 0.065
Chin, Shirvin, Hayes, Silver (35)		Air	0.83 to 1.17	0.85 to 1.20	0.25 to 2.5	0.056 to 0.145
Papell, Trout (39)		Air	0.52 ^c to 3.54	0.32 to 1.8	0.0 to 13.9	0.15 to 0.80
Hatch, Papell (11)		He	2.45 to 6.5	0.34 to 0.67	0.018 to 1.55	0.53 to 0.57
Papell (53)	 AT ANGLES 90°, 80°, 45°	Air	1.2 to 4.8	0.37 to 0.85	0.05 to 12.0	0.20 to 0.70
Seban (33)		Air	0.88	1.14	0.17 to 20.8	0.0045 to 0.13
Seban (34)		Air	0.88	1.0 to 1.14	0.27 to 0.76	0.09
Chin, Skirvin, Hayes, Burggraf (38)	 MULTIPLE SLOTS	Air	1.15 to 1.13	0.87 to 0.887	0.0512 to 1.026	0.0822 to 0.152

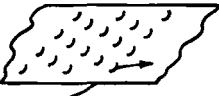
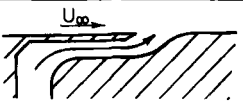
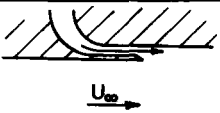
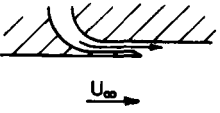
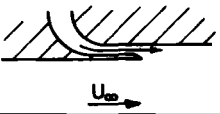
^a Unless otherwise noted, mainstream is air. ^b Tests also for single and double rows of holes. to slot of 10.25 mm and 5.65 mm. ^c Values assume same as Seban (33).

EXPERIMENTAL STUDIES IN FILM COOLING

Velocity ratio, U_s/U_∞	Velocity U_∞ m/sec	Slot size, mm	Starting length Reynolds No., $Re_{x'}$	δ^*/s	Measured parameters, $q/T_{aw}/C_w$	Effec- tiveness	Slot Reynolds No., Re_2	Range of x/s
0.246 to 2.44	15.8 to 32.0	5.0 10.0	8.0×10^5 to 15.0×10^5	0.16 to 0.36	T_{aw}	1.0 to 0.05	360 to 1200	80 to 800
0.286 to 1.35	9.7 to 20.4	3.175	1.2×10^6 to 2.9×10^6	0.22 to 0.33	q T_{aw}	0.60 to 0.095	550 to 2500	10 to 130
0.097 to 1.06	13.0 to 37.0	3.175	1.3×10^6 to 3.8×10^6	0.205 to 0.216	q T_{aw}	1.0 to 0.05	580 to 2600	10 to 130
0.286 to 1.35	9.7 to 20.4	3.175	1.0×10^6 to 2.09×10^6	0.212 to 0.222	q T_{aw}	0.7 to 0.095	550 to 2500	10 to 130
0.26 to 2.85	18.9 to 54.0	2.7	15.0×10^5 to 160.0×10^5	0.94 to 2.36	T_{aw}	1.0 to 0.13	1400 to 8200	9 to 233
	168.0 to 425.0	1.59 3.175 6.35 12.7				1.0 to 0.0	0 to 450,000	0 to 540
0.095 to 3.84	256.0 to 316.0	3.175 6.35 12.7			T_{aw}	1.0 to 0.16	672 to 46,802	3.18 to 271
0.036 to 2.55	147.0 to 395.0	6.35 ^c slot 6.35 holes			T_{aw}	0.95 to 0.22	2,500 to 260,000	4 to 152
0.193 to 23.6	1.52 to 45.8	1.59 3.175 6.35	0.4×10^6 to 1.1×10^6	0.03 to 0.12	q T_{aw}	0.95 to 0.04	620 to 7950	5 to 300
0.31 to 0.865	30.5	1.59 3.175 6.35	0.76×10^6 to 13.4×10^6	0.03 to 1.2	q T_{aw}	0.75 to 0.05	760 ^f to 4400	2 to 300
0.057 to 1.13	30.0 to 56.0	2.92	10.3×10^5 to 19.5×10^5		T_{aw}	1.00 to 0.174	413 to 6100	7.9 to 177.8

^c Estimated. ^d Also considered injection through multiple rows of discrete holes. ^e Holes equivalent

TABLE I

Ref.	Geometry	Injection gas ^a	Density ratio, ρ_2/ρ_∞	Temp. ratio, T_2/T_∞	Blowing rate, $M = \rho_2 U_2/\rho_\infty U_\infty$	Free stream Mach No.
Burggraf, Chin, Hayes (70)		Air	1.15 to 1.13	0.87	0.0861 to 1.25	0.0671 to 0.1712
MULTIPLE LOUVERS						
Hartnett, Birkebak, Eckert (30)		Air	0.78 to 0.98	1.02 to 1.27	0.265 to 0.288	0.145
Hartnett, Birkebak, Eckert (46)		Air	0.875 to 0.935	1.07 to 1.14	0.28 to 1.23	0.1185
Seban, Back (15)		Air	0.90 to 0.97	1.03 to 1.11	4.95 to 12.6	0.00965 to 0.0149
Seban, Back (14)		Air	0.88	1.13	0.0 ^h to 0.70	0.04 to 0.095
Nishiwaki, Hirata, Tsuchida (41)		Air	0.825	1.21	0.0 to 0.0975	0.017 to 0.086
Seban, Back (13)		Air	0.87 to 1.00	1.0 to 1.15	0.2 to 0.9	0.083 to 0.110
Eckert, Birkebak (31)		Air	0.87	1.15	0.19 to 0.93	0.14
Goldstein, Shavit, Chen (3)		Air	0.83 to 0.95	1.05 to 1.207	0.012 to 0.040	0.0961 to 0.1615
Samuel, Joubert (37)		Air	1.11 to 1.28	0.78 to 0.905	0.25 to 3.18	0.040 to 0.085
Goldstein, Eckert, Tsou Haji-Sheikh (63)		Air	3.4 to 2.04	0.8 to 1.25	0.0 to 0.408	3.01
		He	0.3 to 0.4	0.31 to 0.39	0.01 to 0.02	3.01

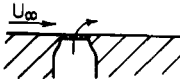
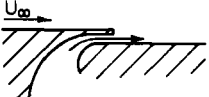
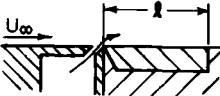
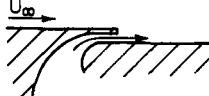
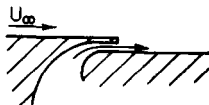
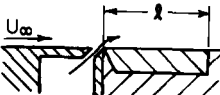
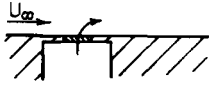
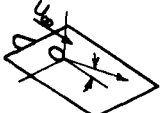
^a Velocity accelerated downstream of injection slot to 2.5 and 1.6 times initial values. ^h Values measured velocity data.

EXPERIMENTAL STUDIES IN FILM COOLING (Continued)

Velocity ratio, U_2/U_∞	Velocity U_∞ m/sec	Slot size, mm	Starting length Reynolds No., Re_x	δ^*/s	Measured parameters, $q/T_{aw}/C_w$	Effec- tiveness	Slot Reynolds No., Re_2	Range of x/s
0.099 to 1.44	25.2 to 52.0	1.62	8.62×10^5 to 18.0×10^5		T_{aw}	0.999 to 0.139	530 to 7700	6.6 to 390.7
0.294 to 0.333	50.0	3.12	6.1×10^5	0.244	T_{aw} q	0.85 to 0.125	1510 to 2880	4 to 140
0.31 to 1.37	42.0	3.11	4.97×10^5	0.2	T_{aw}	0.96 to 0.19	2,200 to 10,000	6 to 198
5.1 to 14.0	3.4 to 5.2	1.59 3.175 6.35			T_{aw} q	1.0 to 0.2	3530 to 6960	17 to 300
0.0 ^a to 0.795	15.8 to 29.5	1.59 3.175 6.35	0.41×10^5 to 0.76×10^5	0.3 ^b to 1.2	q T_{aw}	0.95 to 0.06	0 to 6600	10 to 150
0.0 to 0.118	6.0 to 30.0	5.0 20.0 50.0			T_{aw}	0.7 to 0.035	0 to 2200	1.4 to 180
0.20 to 1.03	29.0 to 38.0	1.59 3.175 6.35	3.3×10^5 to 16.8×10^5	0.24 to 1.36	T_{aw}	0.95 to 0.06	1000 to 7000	5 to 280
0.218 to 1.07	50.5	3.2	6.5×10^5	0.2	T_{aw}	1.0 to 0.10	1800 to 8500	2 to 275
0.013 to 0.042	30.5 to 55.0	35.6	0.59×10^5 to 1.06×10^5	0.0359 to 0.0403	T_{aw}	0.85 to 0.05	850 to 5800	1.5 to 17.44
0.23 to 2.48	15.2 to 30.5	3.175 6.35 9.525			T_{aw}	1.0 to 0.2	1420 to 22,900	3.6 to 275.0
	1050	1.63 3.12 4.62	4.0×10^5	0.045 to 0.127	T_{aw}	1.0 to 0.1	0 to 11,100	0 ^c to 73
	1050	1.63	4.0×10^5	0.127	T_{aw}	1.0 to 0.1	0 to 231	0 ^c to 73

taken from Seban (33). ^a Based on total step height. ^b Based on tunnel dimension rather than

TABLE I

Ref.	Geometry	Injection gas ^a	Density ratio, ρ_2/ρ_∞	Temp. ratio, T_2/T_∞	Blowing rate, $M = \rho_2 U_2/\rho_\infty U_\infty$	Free stream Mach No.
Mabuchi (43)		Air	0.788 to 0.878	1.14 to 1.27	0.02 to 0.146	0.02 to 0.031
Goldstein, Rask, Eckert (42)		Air	0.84 to 0.88	1.12 to 1.20	0.013 to 0.052	0.095 to 0.16
		He	0.12	1.12 to 1.20	0.0022 to 0.0076	0.095 to 0.16
Nicoll, Whitelaw (18)		Air with He tracer	1.0	1.0	0.47 to 2.26	0.06
Metzger, Carper, Swank (54)		Air	0.96 ^m	1.05 ^m	0.25 to 1.49	0.04 to 0.07
Kacker, Whitelaw (36)		Air with He tracer	1.0	1.0	0.3 to 2.1	0.06
Whitelaw (40)		Air with He tracer	1.0	1.0	0.47 to 2.24	0.06
Carlson, Talmor (49)		N ^p	2.76	0.363	0.5 to 1.98	0.1 to 0.5
		N ^p	2.76	0.363	0.563 to 2.66	0.115 to 0.242
Escudier, Whitelaw (47)		Air	1.0	1.0	0.023 to 0.074	0.0702
Goldstein, Eckert, Wilson (24)		Air	0.33 to 0.48	0.77 to 1.14	0.0085 to 0.0223	2.9
Goldstein, Eckert, Ramsey, (71) and (72)		Air	0.84	1.2	0.10 to 2.0	0.08 to 0.17

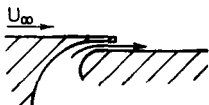
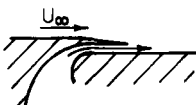
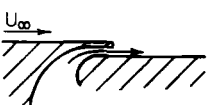
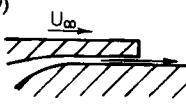
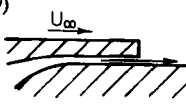
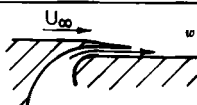
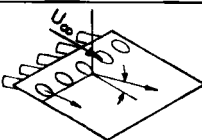
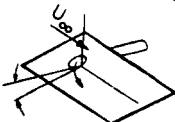
^k From Whitelaw (40). ^l As temperatures are presented for only one case, assumed same for all given boundary thickness. ^o Nitrogen used as mainstream gas. ^p Laminar boundary layer ahead of

EXPERIMENTAL STUDIES IN FILM COOLING (Continued)

Velocity ratio, U_2/U_∞	Velocity U_∞ m/sec	Slot size, mm	Starting length Reynolds No., Re_x , δ^*/s	Measured parameters, $q/T_{aw}/C_w$	Effectiveness	Slot Reynolds No., Re_2	Range of x/s	
0.26 to 0.185	7.0 to 10.0	15.1 to 51.0	8.0×10^5 to 22.0×10^5	0.011 to 0.092	T_{aw}	0.95 to 0.06	358 to 1320	1 to 35
0.015 to 0.0605	33.2 to 55.0	2.54	8.0×10^5 to 14.0×10^5	0.037 to 0.046	T_{aw}	0.80 to 0.04	700 to 4000	0.78 to 31.0
0.018 to 0.063	33.2 to 55.0	2.54	8.0×10^5 to 14.0×10^5	0.037 to 0.046	T_{aw}	0.80 to 0.03	150 to 300	0.78 to 31.0
0.47 to 2.26	21.4	6.425	2.4×10^5 ^k	0.107	C_w	0.95 to 0.20	4,035 to 19,500	4 to 218
0.26 to 1.55	15.2 to 24.4	0.907 to 2.54			q ⁿ T_{aw}	"	312 to 2420	$l/s = 35$ to 70
0.3 to 2.1	21.4	1.87	1.8×10^5 ^q to 11.0×10^5	0.3 to 1.28	C_w	0.7 to 0.1	730 to 5000	50 to 200
0.47 to 2.24	21.2	6.425 to 7.25	2.4×10^5 ^k	0.095 to 0.107	C_w	1.0 to 0.26	4,035 to 19,500	4 to 218
0.17 to 0.67		1.59			T_{aw}	1.0 to 0.11	10,350 to 565,000	
0.19 to 0.90		1.59			T_{aw}	1.0 to 0.11	13,300 to 368,000	
0.023 to 0.074	24.4	25.4			C_w	1.0 to 0.1	958 to 3000	0.625 to 20.6
	1025	12.6	6.0×10^5 ^q	0.026	T_{aw}	0.9 to 0.12	1200 to 3700	0.25 to 8.0
0.12 to 2.38	30.5 to 61.0	23.5 ^r	6.0×10^5 to 21.0×10^5	0.033 to 0.058	T_{aw}	0.85 to 0.005	10,000 to 100,000	2.0 to 40.0

runs. ^m Average heat transfer over section from $x = 0$ to $x = l$ was measured. ⁿ Calculated from injection. ^q Hole diameter. ^r From Whitelaw (40).

TABLE I

Ref.	Geometry	Injection gas ^a	Density ratio, ρ_2/ρ_∞	Temp. ratio, T_2/T_∞	Blowing rate, $M = \rho_2 U_2 / \rho_\infty U_\infty$	Free stream Mach No.
Kacker, Whitelaw (50)		Air with He tracer	1.0	1.0	0.288 to 2.66	0.06
Pai Whitelaw (59)		Air H Air with He tracer Argor (Refrig. 12)	0.07 to 4.17	1.0	0.021 ^u to 6.87	>0.13
Kacker, Whitelaw (51)		Air with He tracer	1.0	1.0	0.2 to 2.4	0.055
Burns, Stollery (29)		Arcton 12 (Refrig. 12)	4.17	1.0	2.21 to 16.7	0.017 to 0.050
		He	0.14	1.0	0.071 to 0.236	0.050
		Mixture Air-He or Air-Arcton	4.17 to 0.14	1.0	0.14 to 4.17	0.050
Metzger, Fletcher (45)		Air	0.96	1.05	0.25 to 0.75	0.04 to 0.07
Pai, Whitelaw (48)		Hydrogen Arcton 12	0.069 to 4.17	1.0	0.021 to 6.85	0.03 to 0.06
Williams (58)		Nitrogen ^z	2.38 to 3.07	0.33 to 0.42	0.308 to 2.99	0.04 ^v to 2.5
Goldstein, Eckert, Eriksen, Ramsey (28)		Air	0.85	1.18	0.1 to 2.0	0.088 to 0.176
		Air	0.85	1.18	0.5 to 2.0	0.088

^a Value for δ taken from Whitelaw (40). ^u Based on maximum possible velocity. ^u Average heat gradients. ^w Air-hydrogen combustion products form mainstream gas. ^z Accelerated flow. ^v Hole

EXPERIMENTAL STUDIES IN FILM COOLING (Continued)

Velocity ratio, U_2/U_∞	Velocity U_∞ m/sec	Slot size, mm	Starting length Reynolds No., Re_x'	δ^*/s	Measured parameters, $q/T_{aw}/C_w$	Effectiveness	Slot Reynolds No., Re_2	Range of x/s
0.288 to 2.66	21.4	1.88 to 3.35 6.35, 12.7	2.4×10^5 to 0.366	0.0542 to 0.366	C_w	0.95 to 0.15	745 to 44,000	25 to 150
0.55 to 2.21	10.1 to 20.8	2.54			C_w	1.0 to 0.005	70.8 to 14,250	2.5 to 212.5
0.2 to 2.4	20.8	6.26	2.17×10^5 to 4.35×10^5	0.107 to 0.191	C_w	0.97 to 0.20	1,500 to 18,400	12 to 210
0.53 to 4.00	6.1 to 17.4	1.59	0.3×10^5 to 0.85×10^5	0.236 to 0.28	C_w	1.0 to 0.3	2,220 to 17,420	0 to 512
0.51 to 1.68	16.8	1.59	0.85×10^5	0.236	C_w	0.9 to 0.05	113 to 368	0 to 512
1.0	17.6	1.59	0.85×10^5 to 3.03×10^5	0.236 to 0.66	C_w	1.0 to 0.05	228 to 11,300	0 to 512
0.26 to 0.78	15.2 to 24.4	1.27 to 2.54			q T_{aw} v		325 to 3500	$l/s = 5.0$ to 70.0
0.55 to 2.21	10.0 to 20.7	2.54			C_w	1.0 to 0.01	70.8 to 14,400	0 to 212
0.127 to 1.09	38.8	0.635 to 1.522			T_{aw}	1.0 to 0.16	3,100 to 24,100	12.7 to 138
0.118 to 2.36	30.5 to 61.0	11.8 z	9.7×10^5 to 17.0×10^5	0.052 to 0.125	T_{aw}	0.85 to 0.0	5200 to 52000	3 to 80
0.588 to 2.36	30.5	11.8 z	9.7×10^5	0.052	T_{aw}	0.4 to 0.0	12900 to 51800	3 to 80

transfer over section from $x = 0$ to $x = l$ was measured. v Favorable and nonfavorable pressure diameter.

At large blowing rates several different empirical and semiempirical relations were used to approximate the wall-jet-like effects observed.

Seban also investigated (34) the influence of mainstream boundary layer thickness at the point of injection through a tangential slot. Only a very slight decrease in film cooling effectiveness was found with increased boundary layer thickness at the point of injection [cf. References (29, 35, and 36)].

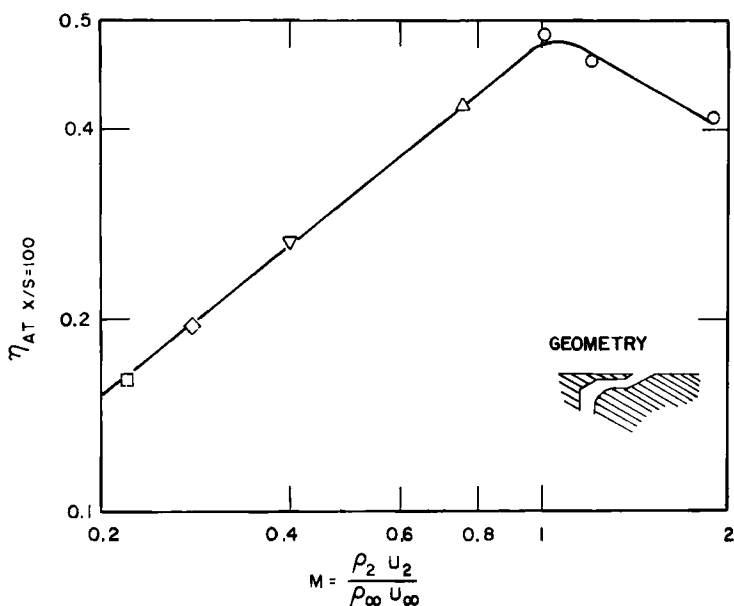


FIG. 8. Film cooling effectiveness at $x/s = 100$ as a function of blowing rate M . [K. Wieghardt, AAF Translation No. F-TS-919-RE (1946).]

Several investigators have used stepdown slots and correlated their data with (different) empirical relations (35, 37); others have also studied the effect of multiple slots (38). Papell and Trout (39) using tangential injection measured the film cooling effectiveness at very large temperature differences. Papell and Trout correlated their results with empirical and semiempirical correlations. Whitelaw (40) measured the impermeable wall concentration for air injection with helium as a tracer gas. The orders of magnitude of the results were found to be similar to previous film cooling effectiveness measurements. The possibility of turbulent Lewis numbers different from unity was suggested. This would reduce the value of a direct comparison, but

would still permit impermeable wall tests to suggest trends and give relative results.

Studies have been made of film cooling downstream of a porous section through which air was injected (3, 41-43). Note that with normal blowing the velocity distribution near the porous section is severely affected (Fig. 9). The results of these different investigations agree quite well. A comparison with results for other geometries is shown in Fig. 10.

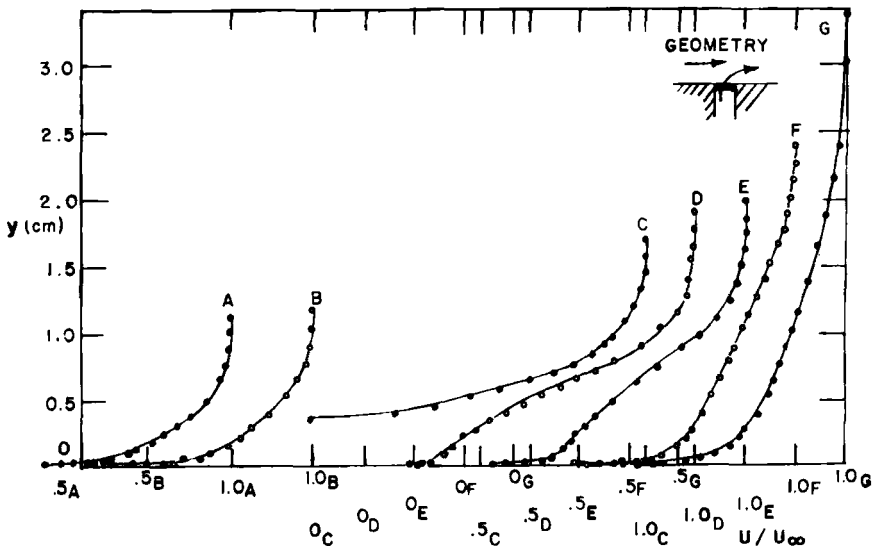


FIG. 9. Effect of injection on boundary-layer velocity profiles with relatively large blowing rate through 35.6 mm porous section with a trip wire 11.2 cm upstream of injection. Data: $\dot{m}_2 = 22.55$ gm/sec, $M = 0.0389$, $U = 56.0$ m/sec. Distance from trip wire (cm): (A) 6.35, (B) 10.16, (C) 15.88, (D) 21.60, (E) 27.95, (F) 53.40, (G) 68.50. [R. J. Goldstein, G. Shavit, and T. S. Chen, *J. Heat Transfer* 87, 353 (1965).]

The film cooling results for porous injection have been found to agree relatively well with the analyses of Librizzi and Cresci (4), Kutateladze and Leont'ev (5), and Stollery and El-Ehwany (6). Actually there is little difference between these three models, and the resulting equations approach the same value far downstream.

Comparisons of the Kutateladze and Leont'ev relations with some of the data for tangential injection are shown in Fig. 11. Agreement is quite good. Figure 12 shows a comparison of some porous wall film cooling experiments with the predictions of Tribus and Klein (Eq. 10), Librizzi and Cresci (Eq. 22), and Goldstein and Haji-Sheikh (Eq. 38). Note the

relatively good agreement with the latter two analyses. In Fig. 13, Eq. (38) is observed to compare favorably with the film cooling effectivenesses obtained for tangential injection. The same relation (Eq. 38) when integrated to predict average values over the length of a wall (44), gives good agreement with the average film cooling effectiveness measured for injection through angled slots by Metzger and Fletcher (45).

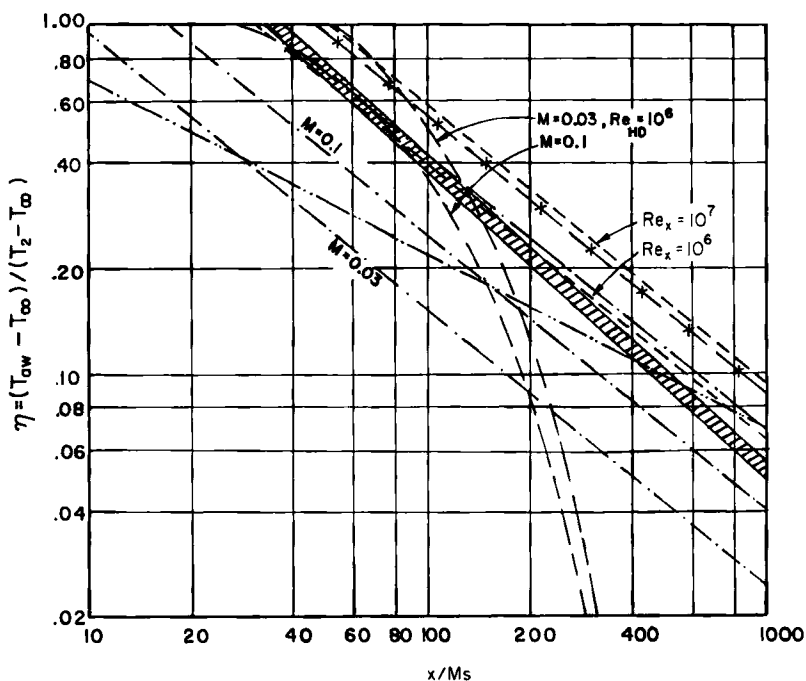


FIG. 10. Comparison of film cooling effectiveness as determined in various investigations.

Data: (|||||), Goldstein *et al.*, (* *) Wieghardt $\eta = 21.8(x/Ms)^{-0.8}$,
 (—) Hartnett *et al.* $\eta = 16.9(x/Ms)^{-0.8}$,
 (---) Nishiwaki *et al.* $\eta = 1.77 Re_x^{0.16} (x/Ms)^{-0.8}$,
 (---) Scesa $\eta = 2.20(x/Ms)^{-0.6}$,
 (---) Seban $\eta = 25.0M^{0.4}(x/Ms)^{-0.8}$,
 (—) Hatch *et al.* $\eta = 1.31 \exp(-0.229 Re_{HD}^{-0.2} (xh^{-1} - 1)/M)$.

[R. J. Goldstein, G. Shavit, and T. S. Chen, *J. Heat Transfer* 87, 353 (1965).]

2. Variable Free Stream Velocity and Free Stream Turbulence

Several investigators (14, 46) have reported studies of film cooling on surfaces with variable mainstream velocity (mainly accelerating). Little change in the boundary-layer temperature profiles was observed

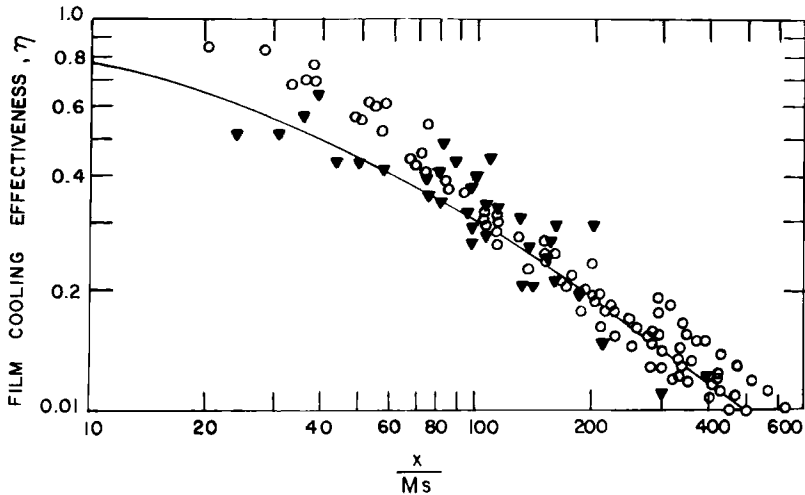


FIG. 11. Comparison of effectiveness for tangential slot geometry with analysis of Kutateladze and Leont'ev. (—) Eq. (25); (○) R. A. Seban, *J. Heat Transfer* **82**, 303 (1960); (▼) S. Papell and A. M. Trout, *NASA Tech. Note TND-9* (1959). [S. S. Kutateladze and A. I. Leont'ev, *Thermal physics of high temperatures* **1**, No. 2, 281–290 (1963).]

and the film cooling effectiveness could be found by multiplying the effectiveness predicted for uniform mainstream flow by a function of the local velocity (46). The relatively small change in effectiveness was attributed (14) to the thermal boundary layer being considerably thicker than the velocity boundary layer. Very strong acceleration (U_∞ increasing by a factor of $2\frac{1}{2}$ or 3) caused a slight decrease in effectiveness.

Escudier and Whitelaw (47) measured the impermeable wall effectiveness for injection through a porous section with strong adverse pressure gradients. Little influence of pressure gradient on effectiveness was observed up to separation, agreeing with the earlier studies in a favorable pressure gradient (45, 46). The small effect observed was an increase in effectiveness. Pai and Whitelaw (48) found little influence of a favorable pressure gradient on impermeable wall effectiveness unless the boundary layer ceases to be fully turbulent.

Carlson and Talmor (49) report a large change (decrease) in film cooling effectiveness with acceleration of the free stream. In their apparatus the test wall along which the secondary gas is injected is not flat; a substantial bend occurs at the point of injection, which may produce separated flow.

They also indicate that increasing the free stream turbulence

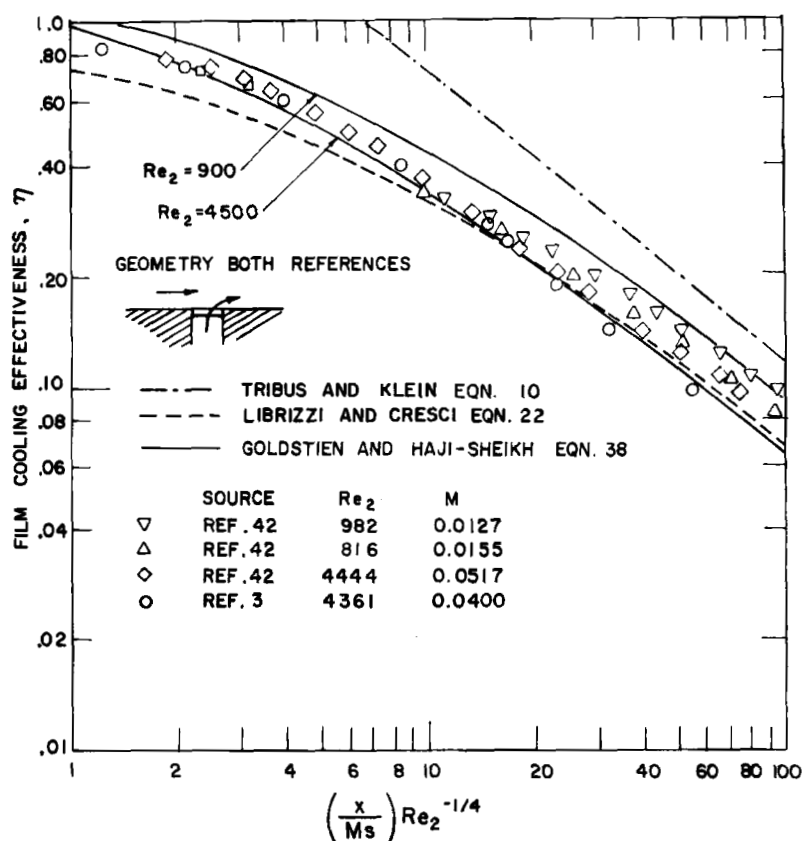


FIG. 12. Film cooling effectiveness with injection of air through a porous section including comparison with several analyses. [R. J. Goldstein and A. Haji-Sheikh, in *Japan Soc. Mech. Engr. 1967 Semi-Intern. Symp.*, pp. 213-218, Tokyo (1967).]

at the slot location can significantly reduce film cooling performance due to the greater mixing of the secondary gas and the free stream. Going from a free stream turbulence intensity of 3 to 22% almost halves the effectiveness some distance downstream of injection. Kacker and Whitelaw (50) changed the turbulence intensity of the secondary gas in the injection slot from 5.5 to 9.5% and found no significant change in impermeable wall effectiveness.

3. Slot Geometry

For tangential injection the ratio of lip thickness t to slot opening s can influence the film cooling effectiveness particularly when the velocity

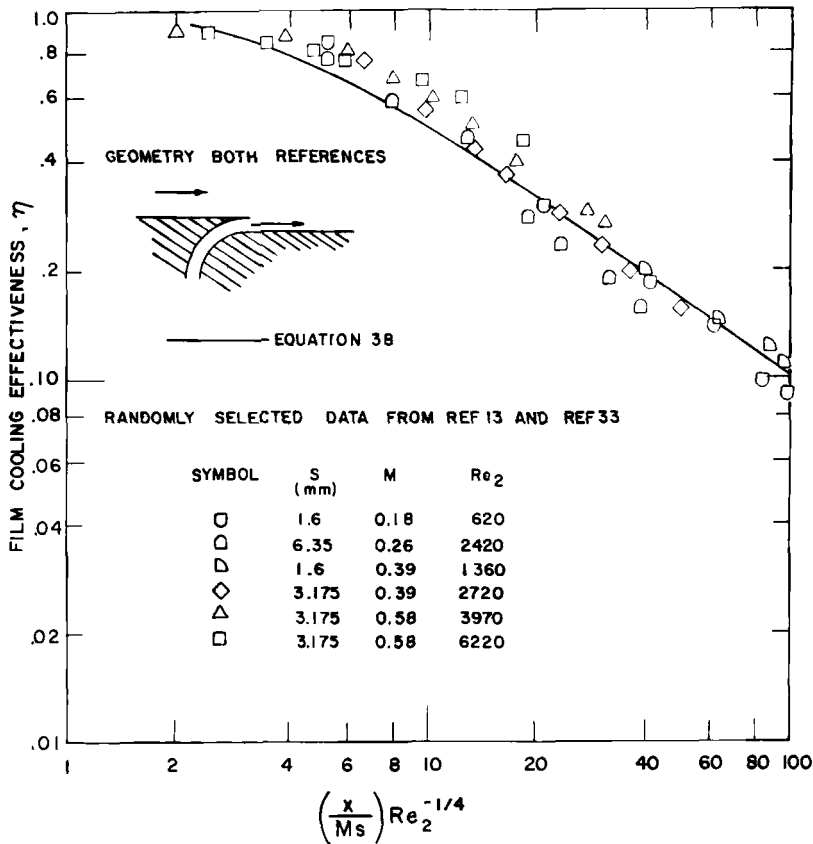


FIG. 13. Film cooling effectiveness with tangential injection including comparison with analysis of Goldstein and Haji-Sheikh. [R. J. Goldstein and A. Haji-Sheikh, in Japan Soc. Mech. Engr. 1967 Semi-Intern. Symp., pp. 213-218, Tokyo (1967).]

ratio U_2/U_∞ is near unity. This has been demonstrated using the mass transfer analogy by Kacker and Whitelaw (50, 51) and Burns and Stollery (29). Figure 14 shows the effectiveness for different values of t/s . For a lip thickness less than about 40% of the slot opening, the effects are small. The influence of lip thickness also diminishes as the velocity ratio U_2/U_∞ is decreased. Similar phenomena are reported by Sivasegaram and Whitelaw (52).

The significant reduction of film cooling effectiveness that occurs for large lip thicknesses is probably due to the pronounced separated and reverse flow region at the lip edge. Under those conditions the simple heat sink models cannot be used directly, though Eq. (25) and

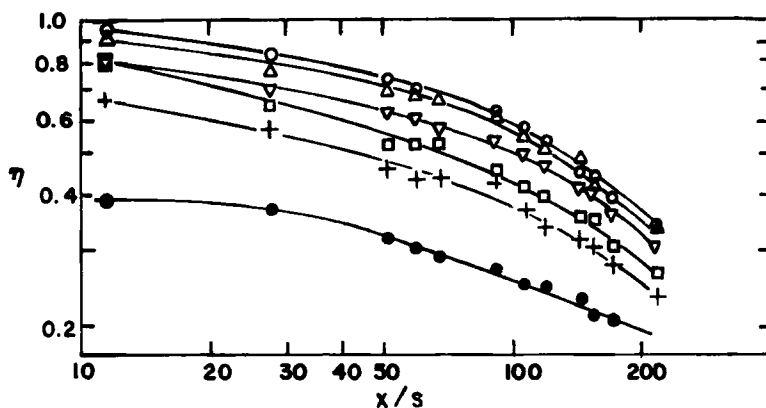


FIG. 14. Effect of increasing slot lip thickness on impermeable wall effectiveness for tangential injection at $p_2/p_\infty \approx 1$ and $U_2/U_\infty \approx 1.07$; t/s : (○) 0.126, (△) 0.38, (▽) 0.63, (□) 0.89, (+) 1.14, (●) 1.90. [S. C. Kacker and J. H. Whitelaw, *J. Mech. Engr. Sci.* 11, 22(1969).]

(38) could possibly be modified to account for the role of geometry in the entrainment of mainstream flow into the boundary layer.

Sivasegaram and Whitelaw (52) report the effect of injection angle on film cooling effectiveness. As expected [cf. Eq. (38)] the larger the angle the smaller the film cooling effectiveness due to the greater mixing of the coolant with the mainstream at the point of injection. Papell (53) and Metzger and co-workers (45, 54) find a similar trend.

4. Effect of Large Temperature Differences

Few experimental studies have used the extreme temperatures that might be encountered in film cooling applications. Large temperature differences can introduce significant errors in assumed boundary conditions and make accurate measurements difficult, particularly in getting adiabatic wall temperature distributions.

Film cooling studies with large temperature differences include those by Papell and Trout (39) and Papell (53) on a flat plate. In their tests the temperature of the hot gas stream was as high as about 800K. Milford and Spiers (55) examined film cooling in a gas turbine combustion chamber at temperatures to 1950K. Lucas and Golladay (56, 57) measured film cooling performance in rocket nozzles and combustion chambers with free stream gas temperatures up to 3000K. Williams (58) studied film cooling in a rocket nozzle with a free stream temperature of about 870K. Other studies made at high temperatures are discussed in the section on compressible flow film cooling.

5. Foreign Gas Injection

There have been few studies of film cooling with the heated or cooled injection of a foreign gas into an air mainstream. Hatch and Papell (11) injected helium through a near tangential slot into a hot air mainstream. Other workers (42) injected heated helium through a porous section. Burns and Stollery (29) find relatively close agreement between these data and a correlation similar to Eq. (31) though the constant is considerably larger, having been increased empirically to give the best fit with experimental data.

The adiabatic wall temperature results for helium injection from (42) were found to be somewhat higher than the prediction of Eq. (21) and (24). However, Fig. 15 shows that Eq. (38) fits the data relatively well.

Investigations with isothermal foreign gas injection have been performed to study the effect of a density difference between the injection gas and the mainstream using the mass transfer analogy. Nicoll and Whitelaw (18) and Burns and Stollery (29) used the mass transfer analogy, injecting foreign gases through tangential slots into an air mainstream and measuring the impermeable wall concentration.

Figure 16 from Reference (29) shows the variation of impermeable wall concentration at a velocity ratio U_2/U_∞ close to unity for different density ratios, ρ_2/ρ_∞ . In this study the influence of boundary layer thickness on effectiveness is found to be small. The thickness of the slot lip plays a significant role near the slot for a relatively light injection gas. A decrease in effectiveness is found with increasing lip thickness which is attributed to increased mixing in the separated region immediately downstream of the lip.

With the heaviest coolant, increasing the velocity ratio U_2/U_∞ increases the effectiveness, though past unity the increase is small. For helium injection the effectiveness continues to increase considerably even for velocity ratios U_2/U_∞ greater than unity.

Pai and Whitelaw (59) measured the impermeable wall effectiveness downstream of a tangential slot through which hydrogen, air (with a helium tracer), argon or Arcton 12 (Refrigerant 12) were injected. With injection of a relatively dense gas the effectiveness reaches a plateau at a velocity ratio U_2/U_∞ of about unity (Fig. 17). For light gas injection increase of the velocity ratio above unity continues to yield further increases in effectiveness. For air injection, the effectiveness at most locations increases with blowing rate, finally reaching a plateau at a velocity ratio U_2/U_∞ of about unity. From unity to the highest velocity ratio used (~ 3.1) the effectiveness stays approximately constant.

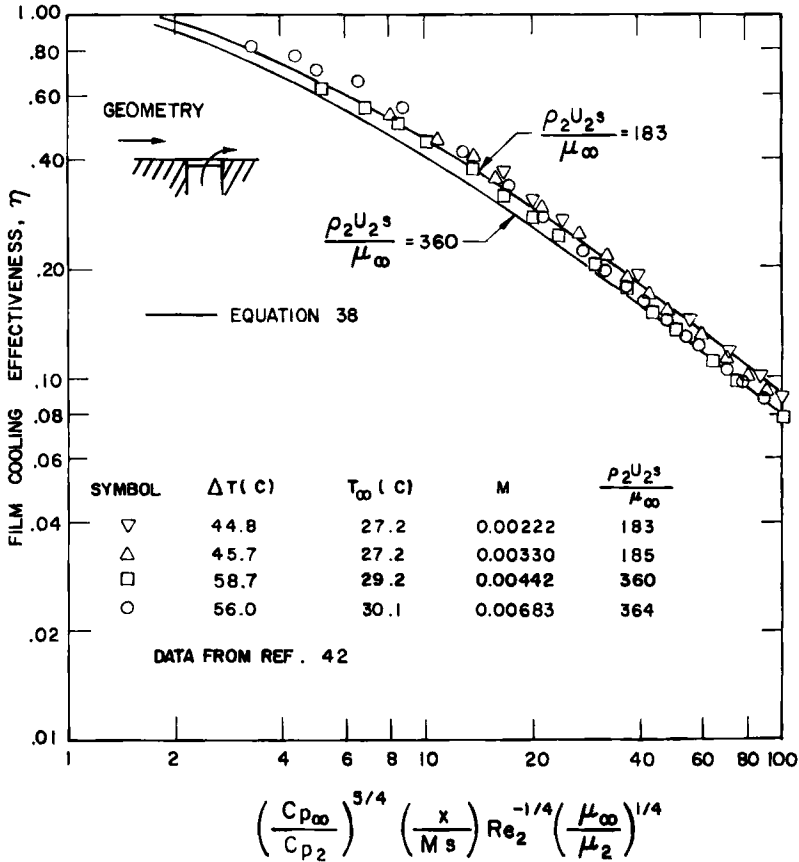


FIG. 15. Film cooling effectiveness with injection of He through a porous section into a mainstream of air including comparison with analysis of Goldstein and Haji-Sheikh. [R. J. Goldstein and A. Haji-Sheikh, in Japan Soc. Mech. Engr. 1967 Semi-Intern. Symp., pp. 213-218, Tokyo (1967).]

It should be noted that the lip thickness t was about 60% of the slot opening in these tests. Their results were in reasonable agreement with calculations made using the turbulent boundary-layer equations.

6. Heat Transfer

Measurements have been made of the heat transfer with film cooling on a surface over which a uniform mainstream flowed (30) and on a surface with a pressure gradient (46). Except at large blowing rates, they reported that shortly downstream of injection the heat transfer coefficient

reduces to the heat transfer coefficient one would expect with no blowing (Fig. 18). Near the injection region the blowing usually causes a slight increase in heat transfer coefficient. The driving force in defining the heat transfer coefficient is the difference between the actual wall temperature and the adiabatic wall temperature. Scesa (60) and Seban and co-workers (14, 32, 33) found similar results in that the heat transfer coefficient was not significantly altered by blowing, although in these studies the heat transfer coefficient was sometimes found to be reduced slightly by the blowing. The difference in injection geometry used in (30) as compared to that used in (33) may account for this different trend. Metzger and co-workers (45, 54) observed a slightly larger effect (increase) on heat transfer coefficient than the other studies, particularly at large blowing rates and close to the injection location.

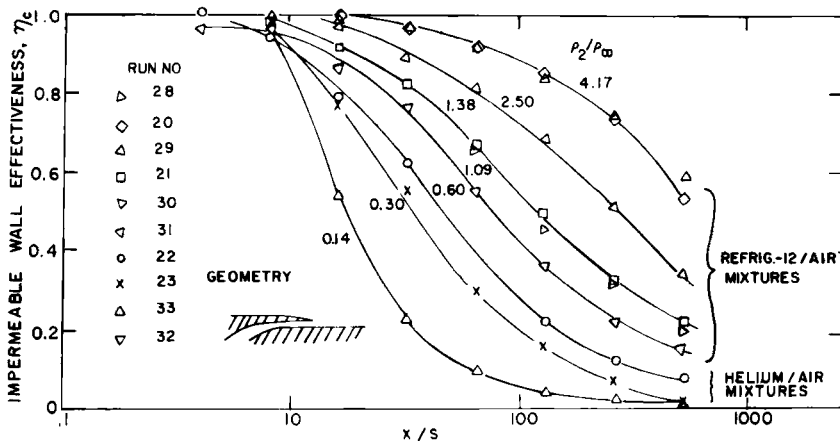


FIG. 16. Effect of density ratio, ρ_2/ρ_∞ , at a velocity ratio $U_2/U_\infty \cong 1$ on impermeable wall effectiveness for tangential injection. [W. K. Burns and J. L. Stollery, *Intern. J. Heat Mass Transfer* 12, 935 (1969).]

C. TWO-DIMENSIONAL FILM COOLING COMPRESSIBLE FLOW

Many applications of film cooling occur in high-speed flows. Although the incompressible flow results can often be used for compressible flow problems, this transformation must be checked experimentally. This is particularly true if the wall geometry is such as to produce shock interactions in the film cooled region.

In several reports (61–63) measurements have been made of the adiabatic wall temperature distribution downstream of a step-down slot in supersonic flow. Either air or helium could be injected tangentially into an air mainstream which had a Mach number of approximately

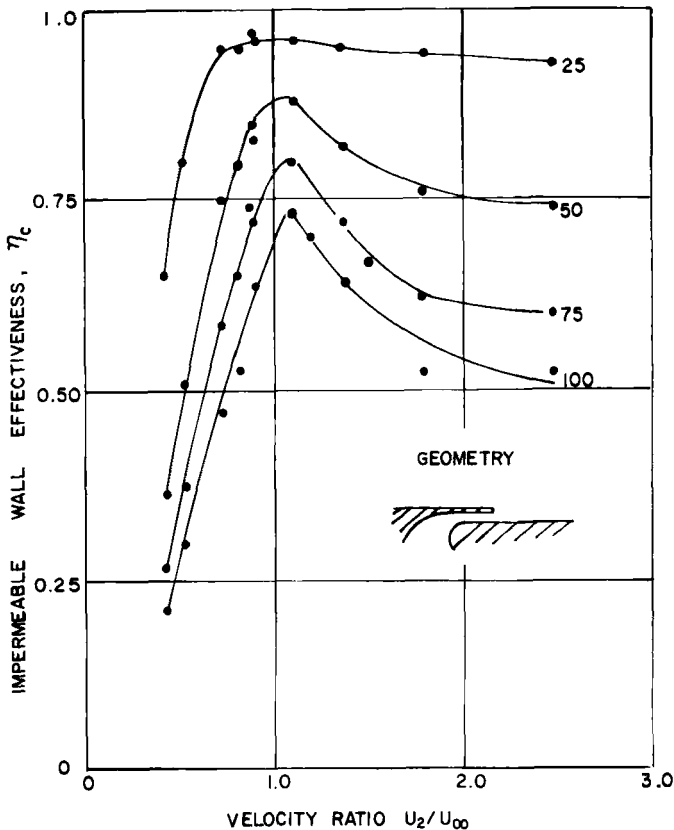


FIG. 17. Impermeable wall effectiveness for air injection with He tracer as a function of velocity ratio U_2/U_∞ . [S. C. Kacker and J. H. Whitelaw, *J. Heat Transfer* 90, 469 (1968).]

three. Both heated and cooled secondary flows were used. Due to the flow over the edge of the splitter plate (separating the secondary and main-stream flows) there is an expansion fan, a lip shock, a separated region, and a reattachment shock, whose magnitudes are dependent on the rate of secondary mass addition. The effect of blowing rate on the flow field is shown by schlieren photography in Fig. 19a and 19b. At the larger secondary flow rates choking occurs in the injection slot.

The results were correlated, using a film cooling effectiveness based on the isoenergetic flow conditions as described earlier. In measuring the isoenergetic film cooling effectiveness two test runs are required

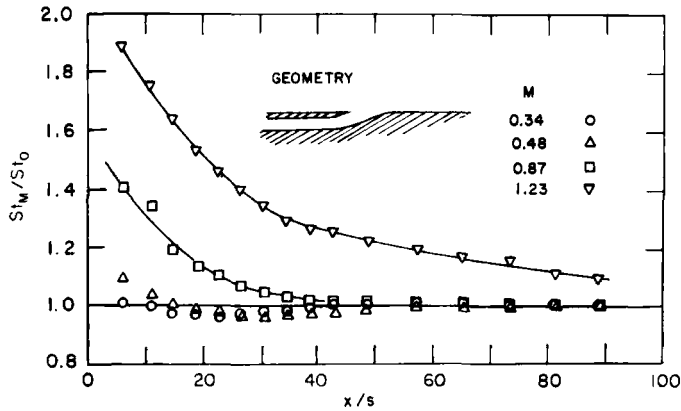


FIG. 18. Ratio of Stanton number with injection St_M to Stanton number for a solid plate with zero injection St_0 for near tangential injection. [J. P. Hartnett, R. C. Birkebak, and E. R. G. Eckert, in "International Developments in Heat Transfer," Part IV, p. 682. ASME, New York (1961).]

at each blowing rate—one to obtain the isoenergetic wall temperature distribution and the other to obtain the film-cooled wall temperature distribution. Since the total temperature of the mainstream may be somewhat different in the two runs it is useful to normalize the temperatures in Eq. (5) by dividing them by the total temperature of the free stream for the test in which they are obtained. An empirical correlation of the results for air injection at low blowing rates (Fig. 20) is

$$\eta_{1s} = 550(\zeta)^{-2.0} \quad \text{for } M < 0.12 \quad (55)$$

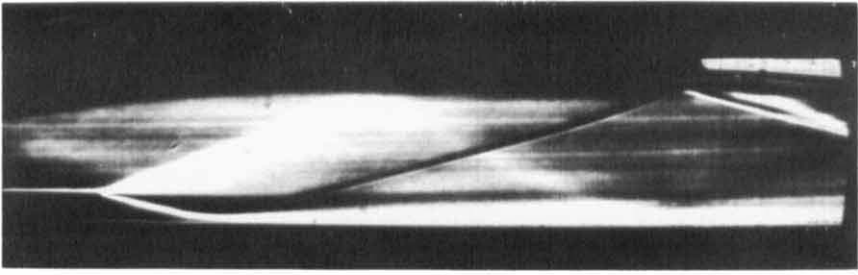
where

$$\zeta = (x/h')[(1/M)]^{0.4} \quad (56)$$

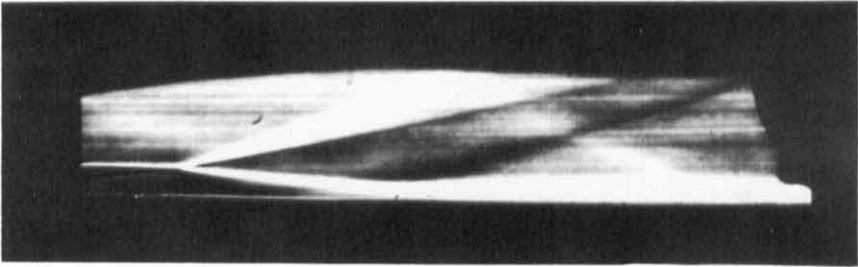
Note that the step height h' (slot height plus lip thickness) is used, indicating the importance of the geometry. These test results are of the order of magnitude of the Tribus and Klein correlation indicating higher values of effectiveness than is found in most of the subsonic studies. At higher values of blowing rate ($M > 0.12$) the effectiveness results are considerably higher than even the Tribus and Klein equation, the empirical correlation over the range of parameter studied being

$$\eta_{1s} = 162(x/Mh')^{-1.2} \quad \text{for } 0.12 < M < 0.408 \quad (57)$$

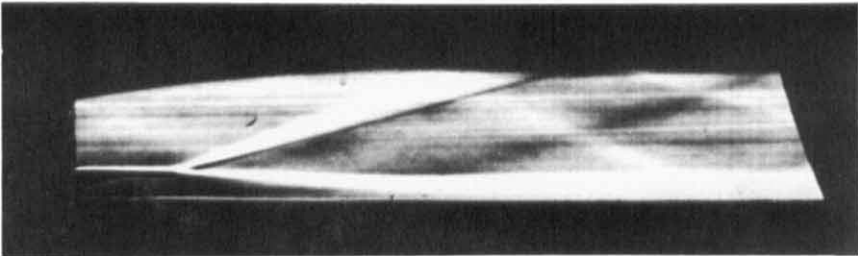
A comparison of these empirical correlations and some of the other predictions is shown in Fig. 21. Note that, compared to subsonic correlations, the supersonic results for injection through a tangential slot



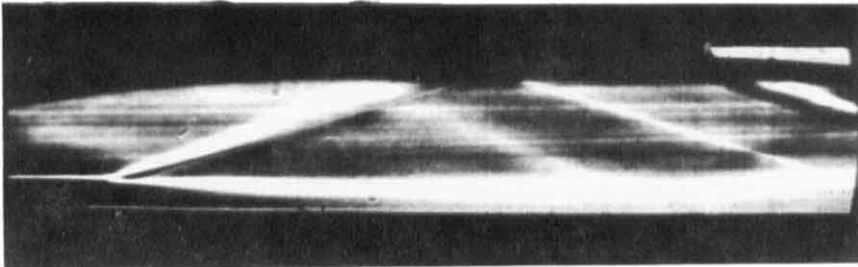
$M = 0$



$M = 0.034$

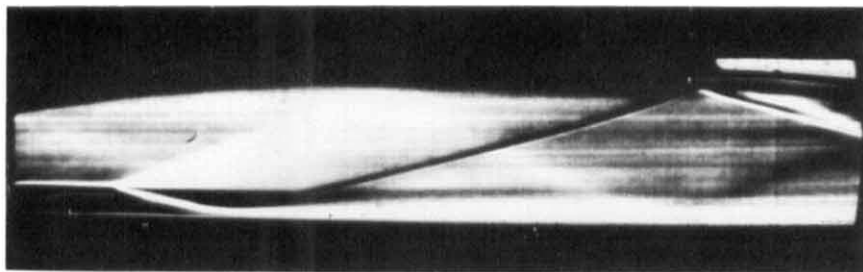


$M = 0.071$

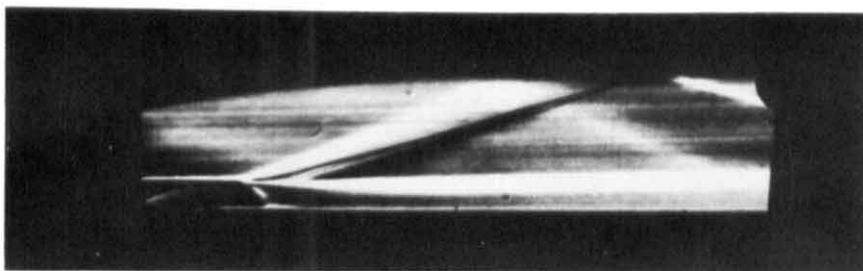


$M = 0.136$

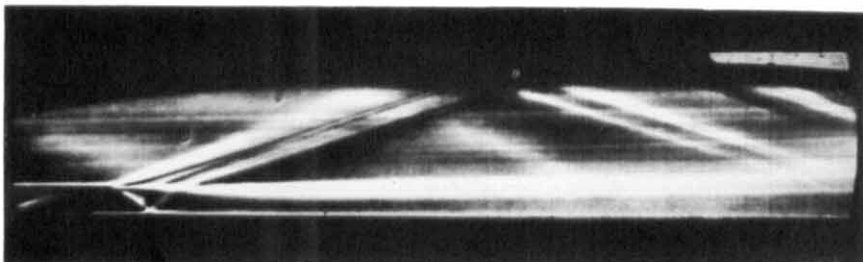
FIG. 19a. Schlieren photographs for injection through a step down slot in supersonic flow for mainstream Mach number = 3.01, slot opening $s = 4.62$ mm, step height $h' = 6.07$ mm. [R. J. Goldstein, E. R. G. Eckert, F. K. Tsou, and A. Haji-Sheikh, *AIAA (Am. Inst. Aeron. Astronaut.) J.* 4, 981 (1966).]



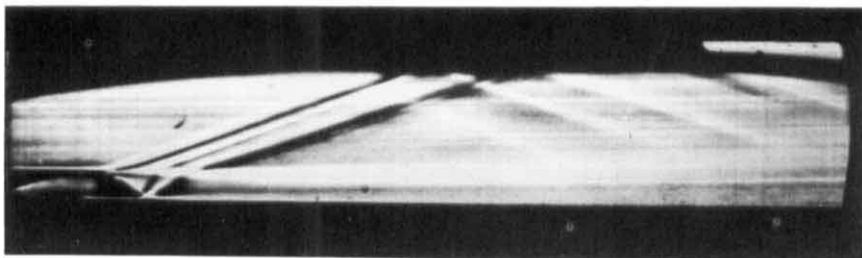
$M = 0$



$M = 0.221$



$M = 0.318$



$M = 0.412$

FIG. 19b. Schlieren photographs for injection through a step down slot in supersonic flow for mainstream Mach number = 3.01, slot opening $s = 4.62$ mm, step height $h' = 6.07$ mm. [R. J. Goldstein, E. R. G. Eckert, F. K. Tsou, and A. Haji-Sheikh, *AIAA (Am. Inst. Aeron. Astronaut.) J.* 4, 981 (1966).]

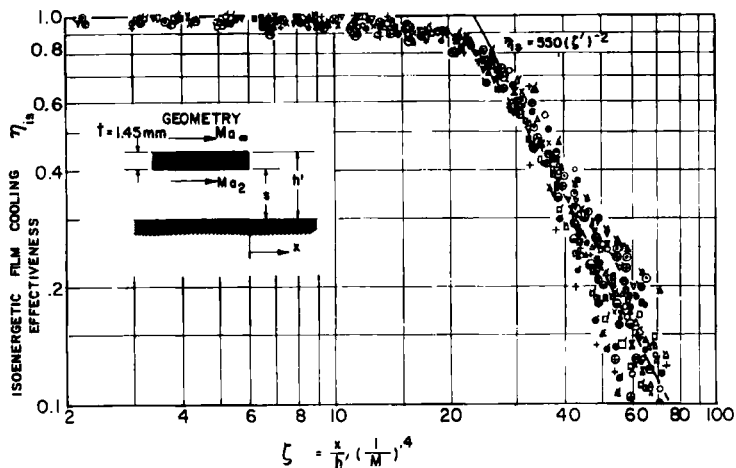


FIG. 20. Correlation of film cooling effectiveness for supersonic mainstream flow with heated air injection at small blowing rates. [R. J. Goldstein, E. R. G. Eckert, F. K. Tsou, and A. Haji-Sheikh, Univ. of Minnesota, Heat Transfer Lab. Rept. HTL TR 60 (1965).]

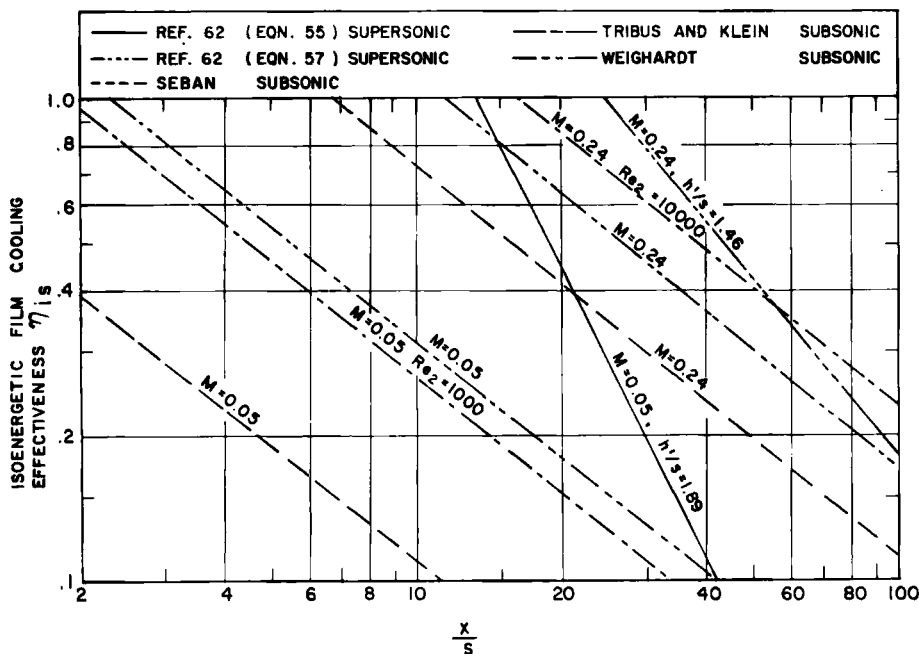


FIG. 21. Film cooling effectiveness as predicted by subsonic and supersonic correlations. [R. J. Goldstein, E. R. G. Eckert, F. K. Tsou, and A. Haji-Sheikh, Univ. of Minnesota, Heat Transfer Lab. Rept. HTL TR 60 (1965).]

indicate a much more substantial length immediately downstream of the slot where the effectiveness is close to unity although further downstream the effectiveness diminishes more rapidly with distance than with subsonic film cooling.

For helium injection through the tangential slot, the film cooling effectiveness based on the recovery temperature (obtained from tests with no blowing) was used (62, 63). The results are correlated by the relation

$$\eta_r = 10,000(x/h')^{-2.0}M^{0.8} \quad \text{for } 0.01 \leq M \leq 0.02 \quad (58)$$

Mukerjee and Martin (64) studied film cooling with injection of air into a Mach 1.5 to 1.7 air mainstream. Their system had approximately tangential injection though the relative lip thickness was much greater than in References (62) and (63). In fact, the slot opening was only from 10 to 30% of the total step height h' . The secondary flow was apparently not increased to the point of choking. They report an empirical relation and compare their measured isoenergetic film cooling effectiveness with the results of Reference (63) and the equation of Tribus and Klein. At low blowing rates relatively good agreement is found with this latter equation though some of the results indicate considerable deviations. The measured effectiveness values are in qualitative accord with the earlier study (63). Differences between the two studies are possibly due to the difference in injection lip geometry. Interestingly, at high blowing rates a significantly shorter length along the wall for which the effectiveness is unity and then a more gradual diminishment of effectiveness with distance is found in Reference (64) as compared to References (62) and (63).

Parthasarathy and Zakkay (65) conducted an extensive series of tests for film cooling with an axisymmetric Mach 6 air mainstream. Helium, hydrogen, argon, and air were employed as coolants with sonic injection in the upstream, normal and downstream directions. The boundary layer thickness at the injection location was much larger than in most other studies. For downstream injection they correlate their results with the relation

$$\eta_0 = K[(x/s) M^{-0.8}]^{-0.7} \quad (59)$$

where $K = 155, 120, 35$, and 30 for injection of hydrogen, helium, air, and argon, respectively. Normal injection gave less effective cooling than injection in the downstream direction, while no correlation could be obtained for injection in the upstream direction. It should be noted that the definition of effectiveness η_0 uses the free stream stagnation temperature as a reference rather than the recovery temperature or isoenergetic temperature.

Dannenberg (66) injected helium nearly tangentially on a hemisphere close to the stagnation point over which a Mach 10 airstream with a stagnation temperature of 4800K was flowing. The peak heating rate was reduced by a factor as large as 2.5 and immediately downstream of the injection point almost complete protection ($\eta_r \approx 1$) could be obtained. The most effective film cooling was found when the velocity of the injected coolant was matched to the velocity of the gas stream at the point of injection.

Film cooling of a Mach 2.4 axisymmetric nozzle was studied by Lieu (67). The mainstream air was heated to 670K. Injection took place near the entrance (subsonic region) to the nozzle through a slot inclined at an angle of 10° to the main flow. Optimum film cooling performance was obtained when the free stream and coolant velocities were approximately equal. A modified version of the Hatch and Papell correlation (11) was used to correlate the results.

Redeker and Miller (68) used film cooling in the stagnation region of a cylinder exposed to a Mach 16 crossflow. Nitrogen and helium could be injected either normal or tangential to the surface. Considerable reduction in heat transfer was found with injection. With normal injection the aerodynamic heating could be cut in half while with tangential injection it could be reduced to one-tenth of the non-film-cooled value.

The downstream film cooling effects of nitrogen, helium, and argon injected through a transpiring flat plate into a Mach 8 airflow was studied by Woodruff and Lorenz (69). The reduction in the turbulent heat flux in the downstream region was found to be relatively independent of Mach number. Use of the blowing parameter eliminated the influence of Reynolds number and the nature of the coolant on the results.

Studies have been made (24) of film cooling with a Mach 3 mainstream and injection of air through a narrow porous strip. Use of a reference temperature enabled them to correlate their adiabatic wall temperatures, in the form of isoenergetic film cooling effectiveness with modified subsonic incompressible flow relations (see section on analysis).

It should be recalled that other film cooling results with compressible flow could not be correlated well with modified incompressible flow correlations. In those, however, the flow geometry was much more complicated. Usually a step down slot of some type was used and the resulting flow pattern, including lip shock, separated region and reattachment shock could be such as to preclude simple correlations.

D. THREE-DIMENSIONAL FILM COOLING

In many applications of film cooling, design considerations prevent the use of continuous slots for introduction of the coolant. Discrete holes may be used for injection, or a slot with discontinuities (due to structural supports) may be used. If the mainstream is essentially two-dimensional in the absence of injection, blowing through discrete openings will result in a nonuniform flow across the span of the film cooled wall. This is the type of three-dimensional film cooling that will be discussed and reviewed in this section. The film cooling effectiveness for an adiabatic wall is still of interest, but now the effectiveness is a function of lateral position as well as downstream distance.

The film cooling effectiveness for injection through discrete holes is usually considerably less than for slot injection at the same rate of secondary flow per unit span. In addition, as the blowing rate M is increased past a relatively low value (perhaps $M \sim 0.5$ for $\rho_2 \approx \rho_\infty$), the effectiveness for injection through discrete holes falls off rapidly. These phenomena can be understood qualitatively by considering the interaction of a nontangential jet and a mainstream. There is usually ample room across the span for mainstream air to flow between the individual secondary flow entrances. At low blowing rates the jets entering the flow are quickly turned toward the surface by the mainstream. As the blowing rate is increased, the jets penetrate into the mainstream permitting mainstream gas to flow around and under the entering secondary flow jets. This separates the injected fluid from the wall and results in relatively low values of film cooling effectiveness. At still higher blowing rates the jets penetrate further and mix more with the mainstream.

It should be noted that the dynamic head or dynamic pressure ratio ($\rho_2 U_2^2 / \rho_\infty U_\infty^2$), rather than the blowing rate M is probably the parameter to use, for a given geometry, in predicting the secondary flow for which significant penetration of the jet (and reduced effectiveness) occurs. The dynamic head ratio would be important in predicting results for an application where the densities of the secondary and mainstream flows are quite different from test results for approximately constant density studies.

Another important parameter would be the geometry of the hole through which the secondary fluid enters. A geometry which turns the secondary fluid (and thus the jet momentum) towards the wall as it enters the mainstream would be desirable in terms of optimizing the film cooling performance.

Wieghardt (10) covered his continuous slot with perforated sheets to study the effects of both a single row and two rows of holes running transversely across the wall. He studied only one blowing rate with this geometry. With two rows of holes the effectiveness was relatively uniform across the span although less by a factor of two than with the same air flow through a continuous slot. With injection through a single row of perforations he found very low values of effectiveness except directly downstream of the central region of each hole.

Papell (53) measured film cooling with injection through multiple rows of discrete circular holes. Injection sections with either two rows or four rows of holes could be inserted in the wall. The holes were at an angle of 90° to the mainflow. Data were taken over a large range of injection rates and could be correlated using an empirical modification of the relation he used for film cooling through a continuous slot.

Use of rows of punched crescent louvers to inject a film coolant has been reported (70). The louvers apparently turn the individual jets downstream so the problem of jet departure from the surface was not

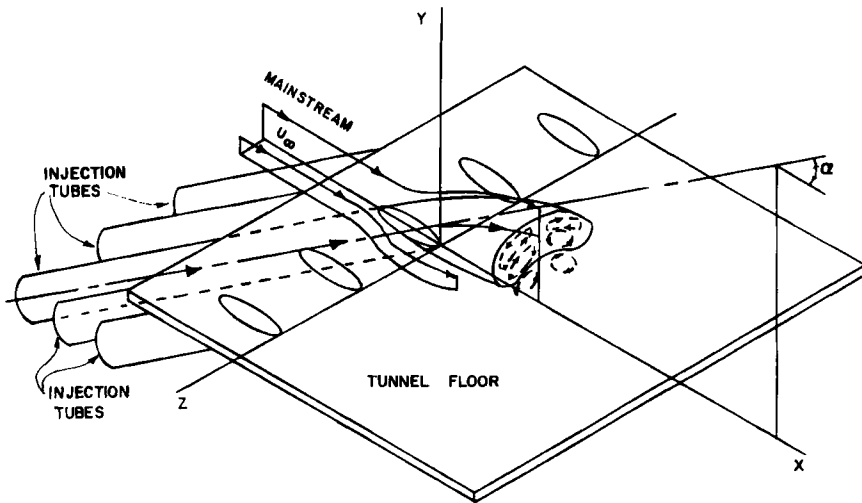


FIG. 22a. Injection section and coordinate system for a row of inclined jets. Detail and flow field are shown for only a single jet interacting with a mainstream. [R. J. Goldstein, E. R. G. Eckert, V. L. Eriksen, and J. W. Ramsey, *Israel J. Technol.* 8, 145 (1970).]

encountered. The data was correlated using the same parameters as were used for film cooling through a number of two-dimensional slots (38). Far downstream the louvers were almost as effective as slots in protecting the surface.

Several publications (28, 71, 72) have appeared from the University of Minnesota on measurements of film cooling with injection through circular tubes (ending flush to the surface) inclined at various angles to the main flow. Both single tubes and a transverse row of tubes were used. The general flow configuration is presented in Fig. 22, which shows qualitatively the flow of the jet entering the mainstream.

Figure 23 shows the film cooling effectiveness downstream of a single hole through which air enters at an angle α of 35° to the main flow. Even along the hole centerline ($Z = 0$) the effectiveness is considerably less than what would be expected for injection through a continuous slot as shown by the top two curves (Eq. 52). Off centerline ($Z \neq 0$)

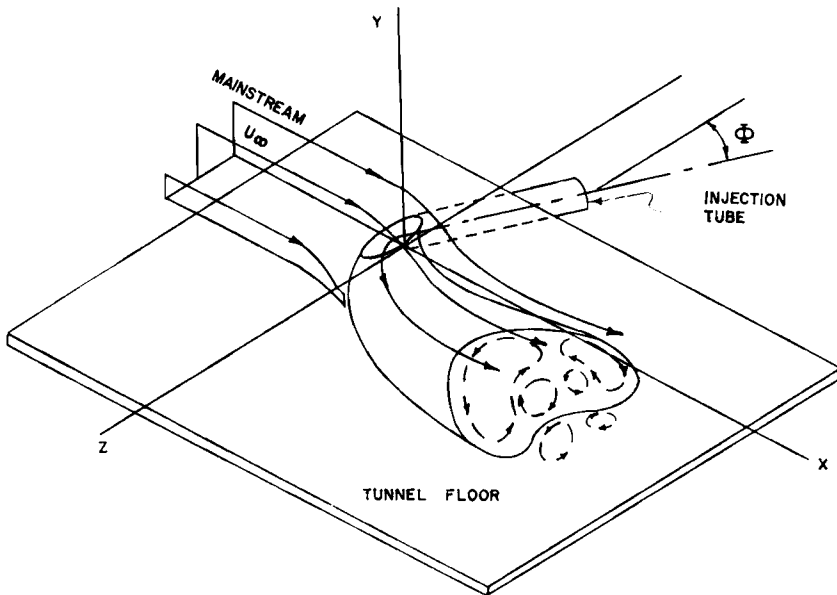


FIG. 22b. Flow field and coordinate system associated with laterally inclined jet interacting with a mainstream. [R. J. Goldstein, E. R. G. Eckert, V. L. Eriksen, and J. W. Ramsey, *Israel J. Technol.* 8, 145 (1970).]

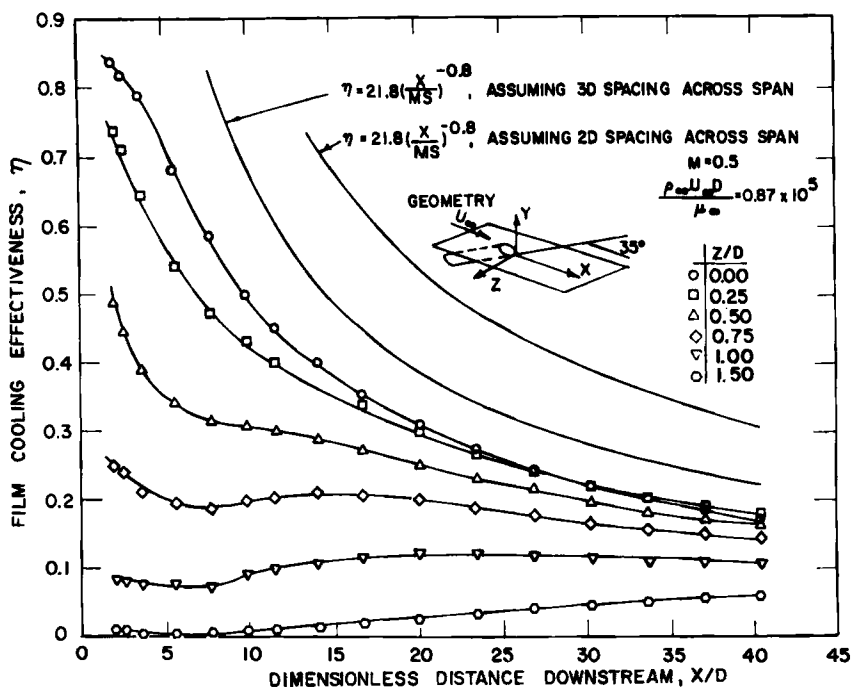


FIG. 23. Axial effectiveness distributions for injection through a single hole at an injection angle of 35° and $M = 0.5$. [R. J. Goldstein, E. R. G. Eckert, and J. W. Ramsey, *J. Eng. Power* 90, 384 (1968).]

the effectiveness is even less. The results shown in this figure are for $M = 0.5$, which is approximately the optimum blowing rate to maximize the film cooling through a single tube at an angle of 35° .

Figure 24 shows how the effectiveness varies with blowing rate at different downstream positions. Data for a single hole and a row of holes inclined at 35° to the mainstream are presented here. Note that for a single row of holes the effectiveness reaches a maximum at a blowing rate $M \cong 0.5$. This could be interpreted as the blowing rate (for $\rho_2 \approx \rho_\infty$) above which the jet is no longer turned by the mainstream to hug the wall along which it enters; above that value it increasingly penetrates into the main flow. At higher blowing rates not only is the effective protection per unit mass of coolant reduced, but the absolute value of effectiveness is reduced as well.

At low blowing rates the flows of the individual jets from a row of holes appear to be independent of one another. The two-dimensional adiabatic wall temperature distribution can then be approximated by

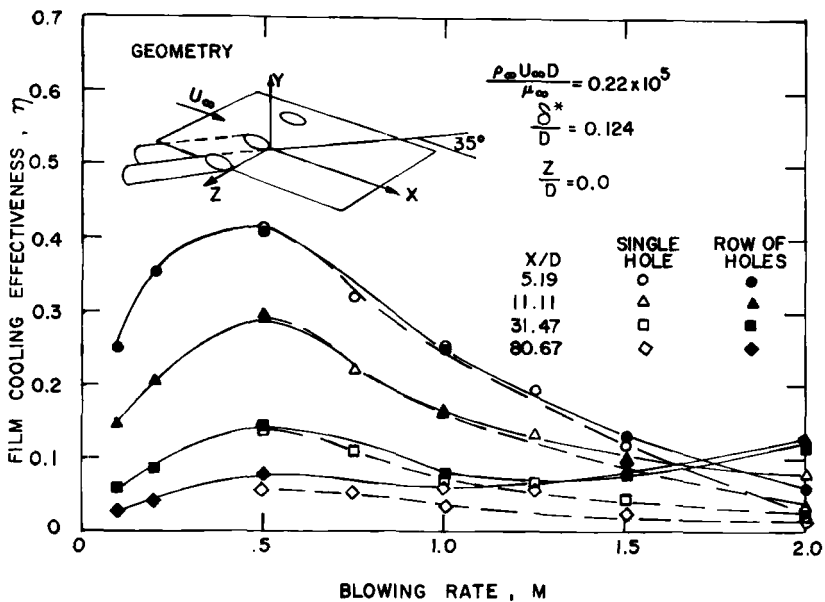
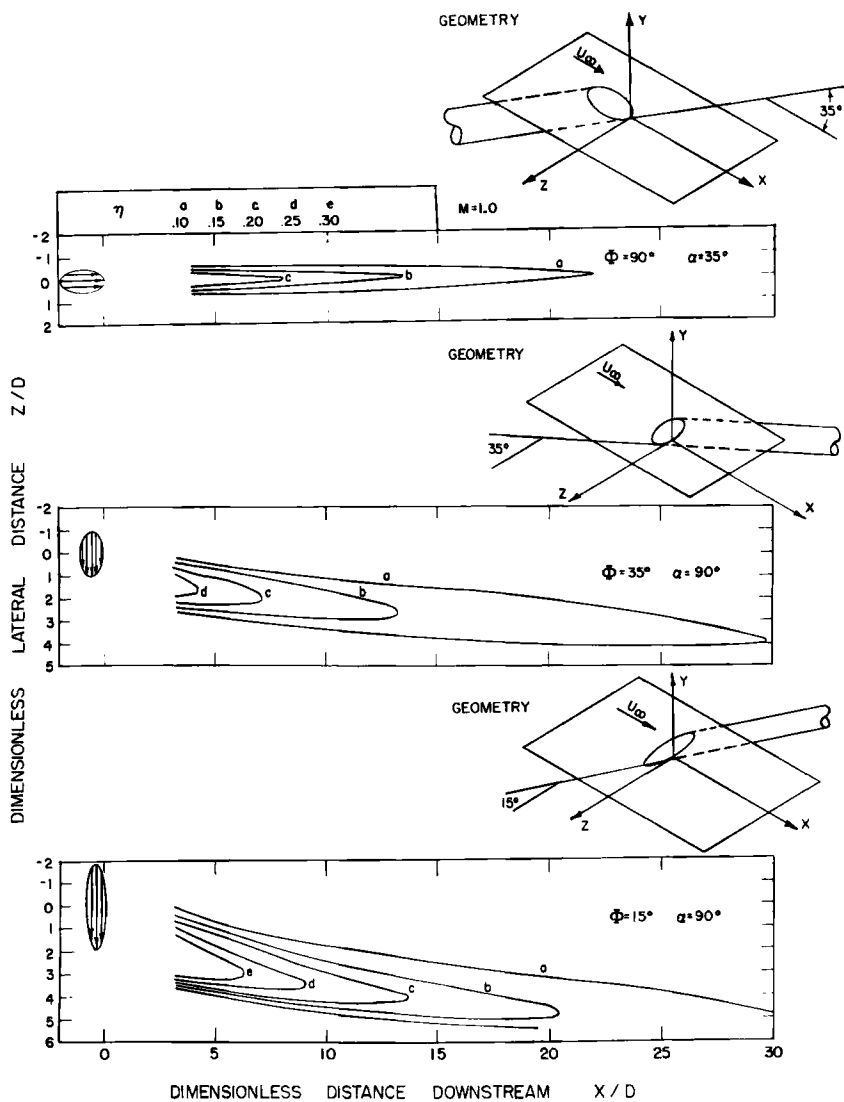


FIG. 24. Comparison of the centerline film cooling effectiveness for single hole and multiple hole injection at an injection angle of 35° with the flow for various blowing rates M . [R. J. Goldstein, E. R. G. Eckert, V. L. Eriksen, and J. W. Ramsey, *Israel J. Technol.* 8, 145 (1970).]

superposition of the effect of a number of single film cooling jets (28). At larger blowing rates where the jets tend to blow off the wall the flows from the individual holes interact. This interaction of the jets results in effectively blocking part of the region where the mainstream might flow around the jets. The secondary flow is then more effectively turned toward the wall, giving considerably higher effectiveness at large M than would be given by a single jet or by superposition (cf. Fig. 24). At large blowing rates the effectiveness for injection through a row of holes increases with position downstream and then remains approximately constant for a considerable distance.

If the injection tube is inclined laterally (Fig. 22b), the film cooling effect is spread out further across the span. At a given blowing rate the average effectiveness across the span can be higher for this geometry than for normal injection or injection through a tube inclined downstream (28). Figure 25 shows contours of film cooling effectiveness for injection through different inclined tubes. Lateral inclination seems to impede the penetration of the jet into the mainstream at moderate values of M .



It should be recalled, however, that M is based on the velocity within the injection tube. As the tube is inclined at a greater angle from the normal the elliptical exit hole area increases and the component of the injection velocity normal to the mainstream decreases, tending to decrease the penetration of the jet into the flow. The effect on penetration is apparently more significant when the other component of the injection velocity is in the lateral direction rather than the downstream direction.

Metzger and Fletcher (45) measured the average film cooling effectiveness (in lateral and downstream directions) following injection through a row of holes inclined downstream. Their trends for the average film cooling effectiveness are similar to those from other studies (71). They also measure the average heat transfer downstream of the holes. Aside from the region close to injection it appears, at least for moderate blowing rates, that the average heat transfer coefficients can be approximated by the values determined without blowing, i.e., for a normal two-dimensional turbulent boundary layer.

V. Concluding Remarks

Considerable understanding of film cooling processes has developed in the last twenty-five years. Recent important applications indicate that there are still significant advances to be made.

Further work on numerical solutions to the equations for turbulent flows should enhance our ability to predict two-dimensional film cooling phenomena. Accurate predictions for film cooling injected at an angle to the mainstream with a relatively thick splitter plate, with high-speed flow or with large density differences may, however, prove elusive.

For secondary flow through discrete holes or even interrupted slots, the difficulties in predicting film cooling performance are even greater. The resulting three-dimensional flow is not yet accessible to anything but simplified analysis. Much work must still be done experimentally to understand the effects of hole geometry, density differences, and the interaction of individual jets on the adiabatic wall temperature distribution. In addition, information on the effect of the mass addition on the local heat transfer is required.

ACKNOWLEDGMENT

Several colleagues were of great aid during the preparation of Table I and in reviewing the manuscript for errors. Particular thanks are due to D. R. Pedersen, who also offered invaluable assistance in preparing the figures and text for publication.

NOMENCLATURE

C	mass fraction of foreign gas	St_M	Stanton number with injection
C_{1w}	mass fraction of foreign gas at an impermeable wall	St_0	Stanton number without secondary injection
C_2	mass fraction of foreign gas present in secondary flow	s	injection slot height
C_∞	mass fraction of foreign gas present in mainstream	t	lip thickness of slot at injection
C_p	specific heat	T_{aw}	adiabatic wall temperature
$\bar{C}_p$	average specific heat of gas in boundary layer, see Eq. (13)	T_{awi}	adiabatic wall temperature with isoenergetic injection
C_{p2}	specific heat of secondary fluid	T_{dat}	datum or reference temperature used in defining the heat transfer coefficient
$C_{p\infty}$	specific heat of mainstream	$T_{0\infty}$	mainstream stagnation temperature
D	diameter of injection tube	T_{02}	stagnation temperature of secondary stream
h	convective heat transfer coefficient	T_{02i}	isoenergetic stagnation temperature of secondary stream (should equal $T_{0\infty}$)
h'	step height, i.e., sum of slot height and lip thickness	T_r	wall recovery temperature in absence of secondary flow
K	empirical constant used in Eq. (59)	T_w	wall temperature
Le	Lewis number; ratio of Schmidt number to Prandtl number	T_{w2}	wall temperature at point of injection, high-speed flow
M	blowing rate or blowing parameter $\rho_2 U_2 / \rho_\infty U_\infty$	T_{w2i}	wall temperature at point of injection with isoenergetic injection
Ma_2	injection Mach number	T_y	temperature at a distance y from the surface
Ma_∞	mainstream Mach number	ΔT	temperature difference
$\dot{m}$	mass flow rate per unit span in boundary layer at any point including both secondary fluid and fluid entrained from mainstream	T_2	temperature of secondary fluid at injection
$\dot{m}_2$	secondary fluid mass flow rate per unit span $\rho_2 U_2 s$	T_∞	mainstream temperature
$\dot{m}_\infty$	mass flow rate per unit span in boundary layer of fluid entrained from mainstream	T^*	property reference temperature
$\dot{m}_{\infty 0}$	mass flow rate per unit span in boundary layer of entrained fluid with no secondary injection	$\bar{T}$	mean temperature in boundary layer, Eq. (12)
Pr	Prandtl number	U	velocity in boundary layer
q	heat flow per unit time and area	U_2	velocity of secondary fluid in injection slot ($\dot{m}_2 / \rho_2 s$)
Re_{HD}	Reynolds number based on hydraulic diameter of tunnel	U_∞	mainstream velocity
Re_x	mainstream Reynolds number based on distance downstream of injection $\rho_\infty U_\infty x / \mu_\infty$	W_2	molecular weight of injection gas
Re_x'	mainstream Reynolds number based on starting length $\rho_\infty U_\infty x' / \mu_\infty$	W_∞	molecular weight of mainstream gas
Re_2	slot Reynolds number based on slot height $\rho_2 U_2 s / \mu_2$	X	distance downstream from point of injection through hole (downstream edge) (Fig. 2)
		x	distance from point of injection (Fig. 1)
		x'	distance from starting position of turbulent boundary layer
		x''	starting length of Reference 5.

y	distance normal to adiabatic wall	η_c	impermeable wall effectiveness, based on concentration, Eq. (6)
Y	distance normal to surface in three-dimensional film cooling studies (Fig. 2)	η_{1a}	isoenergetic film cooling effectiveness, Eq. (5)
$Y_{1/2}$	vertical position at which $(T(X, Y, 0) - T_\infty)/(T(X, 0, 0) - T_\infty) = \frac{1}{2}$	η_0	film cooling effectiveness based on total temperature of gas stream
Z	lateral distance from centerline of injection (cf. Fig. 2)	η_r	film cooling effectiveness based on recovery temperature, Eq. (4)
$Z_{1/2}$	lateral position at which $(T(X, 0, Z) - T_\infty)/(T(X, 0, 0) - T_\infty) = \frac{1}{2}$	λ	dimensionless temperature parameter $(\bar{T} - T_\infty)/(T_{aw} - T_\infty)$
α	angle of injection in YX -plane (Fig. 2)	μ_s	viscosity of secondary fluid
β	injection parameter of Reference 8, Eq. (39)	μ_*	viscosity at reference temperature T_*
δ	boundary layer thickness	μ_∞	viscosity of mainstream fluid
δ_i	boundary layer momentum thickness	ξ	dimensionless film cooling parameter defined in Eq. (9)
δ_T	thermal boundary layer thickness	ξ'	dimensionless film cooling parameter defined in Eq. (40b)
δ^*	boundary layer displacement thickness	ξ_*	dimensionless film cooling parameter for high speed flow defined in Eq. (42)
ϵ	turbulent thermal diffusivity	ρ_s	density of secondary fluid
ζ	parameter defined in Eq. (56)	ρ_*	density at reference temperature T_*
η	film cooling effectiveness, low-speed flow, Eq. (3)	ρ	density of mainstream fluid
		Φ	angle of injection in XZ -plane (Fig. 22)

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